II श्री गणेश प्रसन्न II

CAMPBELL-WALSH-WEIN · BOOK 17

Prostate Cancer

Concise rapid revision · facts answered in Papers 1–4 point to that answer

highlighted number to memorise · KEY POINT and ALGORITHM boxes carry the reasoning · P2 = covered in full in that Paper's answer book; the list under it gives what that answer misses

Chapter	Topic
Ch 153	Epidemiology, Aetiology and Prevention of Prostate Cancer
Ch 154	Prostate Cancer Biomarkers
Ch 155	Prostate Biopsy: Techniques and Imaging
Ch 156	Pathology of Prostatic Neoplasia
Ch 157	Diagnosis and Staging of Prostate Cancer
Ch 158	Active Management Strategies for Localised Prostate Cancer
Ch 159	Active Surveillance of Prostate Cancer
Ch 160	Open Radical Prostatectomy
Ch 161	Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy
Ch 162	Radiation Therapy for Prostate Cancer
Ch 163	Focal Therapy of Prostate Cancer
Ch 164	Treatment of Locally Advanced Prostate Cancer
Ch 165	Management of Recurrent and Newly Metastatic Prostate Cancer
Ch 166	Management of Castration-Resistant Prostate Cancer
★	Putting it together

153 Epidemiology, Aetiology and Prevention of Prostate Cancer

1 · Epidemiology — the headline

- It is the **second commonest** non-skin cancer and the **fifth** cause of cancer death in men; 2020 saw ≈ **1.4 million cases, 375,000 deaths**.
- Incidence varies ~13-fold (South Central Asia lowest, Northern Europe highest). US lifetime risk is ≈ **1 in 8**. 5-year survival is 97%, yet it is the second cause of cancer death in US men.
- **Incidence and mortality are decoupled.** Incidence follows screening and wealth, while mortality peaks where African ancestry meets poor access: the **Caribbean has the highest mortality in the world** MCQ.
- **Latent (autopsy) cancer** is ~5% under 30 y, 29% at 60–69 and **59% >79 y**, and higher in Black men at every age.

> **KEY POINT**
> In a prostate-specific antigen (PSA)-screened population, risk-factor studies may find determinants of being tested, so the preferred end point is **aggressive disease**. **Healthy-user bias**: health-conscious men seek PSA tests, so healthy habits falsely seem to raise incidence while lowering mortality.

2 · Established risk factors

Only three are unequivocal: **age, race (ancestry) and family history**.

- **Age**: mean age at diagnosis is 66 and it is rare under 40. Diagnosis at ≥70 carries a higher risk of cancer death.
- **Race**: Black men have the highest incidence (172.6/100,000 in the US) and **twice** the cancer-specific mortality of White men, presenting younger and more metastatic. Drivers are unequal access, comorbidity and **8q24** (and 17q21) alleles; with **equal access, outcome is equal** in non-metastatic disease.
- **Family history** is present in ~20%. Twin-study heritability is up to **58%**, so genes outweigh shared lifestyle.

Family history	Relative risk
Any first-degree relative (father or brother)	2–3×
Two first-degree relatives	~3.5×
Both father and brother	**9×**
Relative diagnosed <60 vs ≥60	~2.5× vs ~1.6×
Father/brother died of prostate cancer	2× risk of prostate cancer death

3 · Germline genetics

The key inherited genes are **DNA-repair genes**, which predict **aggressive** cancer.

Gene	Pathway	Association	Implication
BRCA2, BRCA1	Homologous recombination (HR)	BRCA2 familial, aggressive	Early screening; poly(ADP-ribose) polymerase (PARP) inhibitor; caution in active surveillance (AS)
ATM	HR	Metastatic disease	As for BRCA
PALB2, CHEK2	HR / DNA repair	Familial; PALB2 lethal	CHEK2: targeted screening
MLH1, MSH2, MSH6, PMS2 (**Lynch**)	Mismatch repair (MMR)	2–5% of cases	Checkpoint inhibitor; very sensitive to androgen deprivation therapy (ADT)
HOXB13	Prostate-specific homeobox	Hereditary prostate cancer	Targeted screening
8q24, 17q21	Genome-wide association study (GWAS) loci	Commoner in African ancestry	Explain part of the racial excess

- **Polygenic risk scores** predict any cancer, not aggressive cancer, so they are not yet practice-changing.

P4 Indications of genomic testing in carcinoma prostate — germline testing: metastatic, high-risk, intraductal/cribriform, family history, Ashkenazi

Not in the paper answer:

- GWAS has found >200 single-nucleotide polymorphisms, each of small effect.

4 · Molecular aetiology and inflammation

- **Chronic inflammation** is in nearly every biopsy and may be causal; **proliferative inflammatory atrophy** is the proposed precursor, via high-grade prostatic intraepithelial neoplasia (HGPIN).
- **TMPRSS2-ERG** fusion puts the ERG oncogene under an androgen-regulated promoter. It is in ~half of Western tumours but uncommon in East Asians. See Ch 154

5 · Possible risk factors — body, metabolism, lifestyle

Most lifestyle factors act on **progression** rather than initiation; evidence is observational.

Factor	Total incidence	Aggressive / fatal
Height	~5% higher per 10 cm	More high-grade cancer and death
Obesity	Mixed; less low-grade	**Consistently raised**
Smoking	Not raised (smokers screen less)	**Mortality raised ~40%**; no excess after quitting ≥10 y
Metabolic syndrome, hypertension, cardiovascular disease (CVD)	Slightly raised	Metabolic syndrome: worse pathology
Diabetes	Lower (low testosterone, IGF-1 and PSA; metformin)	—
Vigorous activity	No clear effect	Less advanced disease

Factor	Total incidence	Aggressive / fatal
Alcohol	Unclear	Mortality raised at ≥45 g/day; binge drinking raises it

- Adiposity raises insulin, IGF-1, leptin, inflammation and oestrogen, which drives proliferation and blocks apoptosis.
- **Adult serum testosterone does not predict risk** MCQ.
- A **composite healthy lifestyle** means **30–50%** less lethal cancer; in the highest polygenic-risk quartile, lifetime lethal risk falls from 5.3% to 1.6%.
- **Diet**: World Cancer Research Fund 2018 found **no dietary factor with strong evidence**; dairy, calcium and lycopene were downgraded.
- **Vasectomy** is not a risk factor (detection bias), and the AUA says it need not be discussed.
- Transgender women on oestrogen have a median PSA ≈ **0.02 ng/mL** (50-fold lower), so cut-points must be lowered.

6 · Supplements

P4 Antioxidants in urological malignancy; Prostate cancer prevention trials — SELECT null, vitamin E harmful; VITAL null; lycopene, green tea, soy/pomegranate/curcumin

Not in the paper answer:

- Calcium supplements raise fatal cancer (relevant on ADT), and zinc ≥75 mg/day raises lethal cancer.
- No supplement benefits healthy men on a balanced diet.

> **KEY POINT**
> Vitamin E worked in the ATBC trial but failed in SELECT: ATBC used a tenfold lower dose in **heavy smokers** with little PSA screening (aggressive cancers); SELECT used 400 IU in non-smokers with widespread screening (indolent cancers). Different disease, different answer.

7 · Chemoprevention

P4 Prostate cancer prevention trials — 5ARI rationale, PCPT/REDUCE high-grade controversy, FDA warning, no mortality effect; aspirin, statins, metformin

Not in the paper answer:

- No preventive agent is FDA-approved. 5α-reductase inhibitors shrink the gland about 25%, so on a 5ARI you **double the PSA**. See Ch 154
- **Secondary prevention**: in the Prostate Cancer Lifestyle Trial (men on AS), plant-based diet, exercise and stress management meant **5% vs 27%** treated at 2 y.
- Activity (≥150 min/week moderate or ≥75 min vigorous) lowers cancer-specific mortality ~31%, while a Western diet and continued smoking raise cancer death.

> **KEY POINT**
> Cardiovascular disease kills more men with non-metastatic prostate cancer than the cancer does, so lifestyle advice benefits both.

8 · PSA screening

P2 PSA screening for prostate cancer: evidence, problems and solutions — trials, corrected ERSPC benefit, overdiagnosis, USPSTF D to C, MRI-first

- Screening lowers cancer death ~**20–30%**, and the benefit builds with time.

9 · Landmark trials

Trial	Design	Result	Take-home
ERSPC (European screening study)	>182,000; PSA every 2–4 y vs none	Cancer death fell 16% (9 y), then ~20% (16 y); excess incidence fell 85% to 36%	Benefit is time-dependent
PLCO (US)	Annual PSA vs usual care	Control arm heavily screened	Contamination dilutes the effect
Göteborg-1	Swedish screening	Significantly fewer cancer deaths	Supports ERSPC
CAP (UK)	Single PSA invitation	No benefit at 10 y; tiny fall in death at 15 y	One-off screening is weak
PCPT	18,882, PSA ≤3; finasteride 5 mg 7 y	Cancer fell **24.8%** (18.4 vs 24.4%); Gleason 7–10 6.4 vs 5.1%; 18-y death fell 25% (not significant)	Fewer cancers, more high-grade
REDUCE	PSA 2.5–10, prior negative biopsy; dutasteride 0.5 mg 4 y	Cancer fell **22.8%**; Gleason 8–10 0.9 vs 0.6%	FDA warning
SELECT	35,533; selenium 200 µg with or without vitamin E 400 IU	Stopped for futility; vitamin E raised risk (P = 0.06)	No supplements
VITAL	>25,000; vitamin D3 2000 IU 5 y	No prostate effect	Not preventive

154 Prostate Cancer Biomarkers

1 · Why biomarkers beyond PSA

- Prostate-specific antigen (PSA) is **organ-specific, not cancer-specific**, which causes negative biopsies and overdiagnosis of Grade Group (GG) 1. Guidelines advise **secondary testing** (imaging or biomarker) before biopsy; the target is **clinically significant cancer (GG ≥2)**. See Ch 153
- Settings are before initial or repeat biopsy, favourable-risk cancer (active surveillance (AS) vs treatment), and after radical prostatectomy (RP).

2 · PSA biology

- PSA is **hK3** (gene KLK3 on **19q13**), an androgen-regulated serine protease that **liquefies semen**. Half-life is **2–3 days**.
- It is made as **proPSA**, and **hK2 cleaves the leader** to give active PSA.
- Serum PSA rises when it leaks through a lost basal layer; per cell, cancer makes less PSA than benign tissue.

Serum form	% of PSA	Measured?
PSA–α1-antichymotrypsin (ACT)	60–90	Yes (predominant, inactive)
PSA–α2-macroglobulin	10–20	**No**: epitopes masked
PSA–α1-protease inhibitor	1–5	Yes
Free PSA	5–45	Yes (proPSA, benign PSA (BPSA), intact PSA)

Confounders

- After biopsy the rise lasts ≥**4 weeks**; ejaculation raises it for 24 h, so repeat after abstinence.
- **5α-reductase inhibitors (5ARIs)** lower PSA ~**50%** by 12 months, so **double** it.
- Age-specific upper limits (ng/mL): 40–49, 2.5; 50–59, 3.5; 60–69, 4.5; 70–79, 6.5.

3 · PSA derivatives

Derivative	Definition / values		Use
PSA density	PSA ÷ volume; cut-off (0.08–0.15)	**0.15**	After negative/equivocal magnetic resonance imaging (MRI); ≤0.15 = Johns Hopkins AS entry; predicts AS reclassification
PSA velocity	Change per year		Minimal value; **never an isolated biopsy trigger**

Derivative	Definition / values	Use
% free PSA	Cancer PSA binds ACT, so **low %fPSA suggests cancer**; thresholds 15–25%	FDA: PSA 4–10, benign digital rectal examination (DRE); mainly **after negative biopsy**; unaffected by 5ARI

> **KEY POINT**
> %fPSA was validated against any cancer, not GG ≥2, so it overestimates significant risk.

- **[-2]proPSA** rises with Gleason and underpins the Prostate Health Index (PHI); **BPSA** tracks benign hyperplasia and has no prognostic value.

4 · Blood and urine tests (NCCN 2023 pre-biopsy)

Six tests estimate GG ≥2 risk; **thresholds are setting-specific** (higher for repeat biopsy).

Test	Components	Threshold / key fact
PHI	Blood: **(p2PSA ÷ fPSA) × √tPSA**	24–28.6 proposed; ~30% biopsies avoided
4Kscore	Blood: tPSA, fPSA, iPSA and **hK2**, with age, DRE, prior biopsy	**7.5%**: sensitivity 94%, negative predictive value (NPV) 95%; ≤7.5% = low long-term metastasis risk
IsoPSA	Blood: PSA isoform structure	Unaffected by 5ARI or α-blockers
SelectMDx	Post-DRE urine DLX1, HOXC6 mRNA with clinical data	NPV ~95%; needs prostate volume
ExoDx (EPI)	Urine exosomes: PCA3, ERG, SPDEF	**No DRE**; 15.6 initial, 20 repeat
MyProstateScore	Post-DRE urine PCA3, TMPRSS2:ERG and PSA	10 initial (sensitivity 97%), 15 repeat; validated in Black men

- **PCA3** is a long non-coding RNA. Score = (PCA3 ÷ KLK3 mRNA) × 1000 after an **attentive DRE**; thresholds 25/35. It is **independent of prostate size** and FDA-approved after a **previous negative biopsy**.
- **TMPRSS2:ERG** is **~100% specific** but in only ~half of tumours, so it is combined with PCA3.

> **KEY POINT**
> Urine tests follow a DRE because cancer RNA comes only from shed cells massaged into the urethra; ExoDx avoids this, as exosomes are shed spontaneously. Urinary markers are highly specific, but their coverage is limited, hence panels.

5 · Tissue-based biomarkers

P2 Molecular staging of CA prostate — Decipher, Oncotype GPS, Prolaris, PORTOS, ConfirmMDx; limits: heterogeneity, retrospective, G-MAJOR

Not in the paper answer:

- **ConfirmMDx** measures GSTP1, APC and RASSF1 methylation in negative cores (NPV ~88–90%), using the **field ("halo") effect** around a missed tumour.
- **Decipher ≥0.4 post-RP** suggests benefit from radiotherapy.

6 · Germline testing

Up to 40–50% of risk is familial, mostly via **DNA-damage-repair** genes.

Gene	Risk	Phenotype
BRCA2	4.5× <65 y; ~8.6× overall	Younger, higher grade, shorter survival
BRCA1	1.8× <65 y	—
ATM, CHEK2	CHEK2 odds ratio ~2–3.4	ATM aggressive
Lynch (MSH2 etc.)	MSH2 ~10×	Also colorectal, upper tract

Gene	Risk	Phenotype
HOXB13 G84E	4.5×	**Not** more aggressive

Testing is for BRCA1/2 alone or a panel (BRCA1/2, ATM, CHEK2, MLH1, MSH2, TP53, EPCAM). Indications: see Ch 153.

7 · Clinical algorithm

ALGORITHM

1. **Raised PSA**: repeat it, excluding infection, instrumentation, ejaculation and 5ARI. Then do MRI and/or a biomarker (PHI, 4Kscore, IsoPSA, SelectMDx, EPI, MPS; PSA density). If risk is high, biopsy.
2. **Negative biopsy, persistent suspicion**: MRI, %fPSA, PCA3/EPI/MPS (repeat thresholds), ConfirmMDx. If positive, targeted re-biopsy.
3. **Favourable-risk**: PSA density and Prolaris/Oncotype/Decipher decide AS vs treatment.
4. **After RP**: PSA should be undetectable. If ≥0.2 ng/mL and confirmed, this is biochemical recurrence; Decipher/PORTOS then guides adjuvant or salvage radiotherapy.

155 Prostate Biopsy: Techniques and Imaging

1 · Indications and contraindications

On transrectal ultrasound (TRUS) the peripheral (PZ) and central zones are homogeneous and echogenic; the transition zone (TZ) is heterogeneous. **39%** of cancers are isoechoic, so a normal grey scale never precludes biopsy.

P2 TRUS / prostate biopsy — indications, contraindications, preparation, technique, TR vs TP, complication rates

Not in the paper answer:

- **Repeat** biopsy follows rising prostate-specific antigen (PSA), atypical small acinar proliferation (ASAP; **≈50%** cancer risk), positive PCA3 or ConfirmMDx, suspicious MRI, or active surveillance (AS).
- National Comprehensive Cancer Network (NCCN): biopsy for abnormal digital rectal examination (DRE) with raised PSA, or repeat PSA >3 ng/mL with PI-RADS ≥3; PSA velocity alone, or abnormal DRE with normal PSA, is not enough. See Ch 154
- PSA density **>0.15 ng/mL/cc** suggests cancer. Volume = π/6 × transverse × anteroposterior × length.

2 · Pre-biopsy mpMRI

P1 P4 Prostate multiparametric MRI; PI-RADS and MRI–ultrasound fusion biopsy — sequences, PI-RADS scoring, zone rules, cancer rate per score, targeting methods

Not in the paper answer:

- Multiparametric MRI is a **triage test**: with no lesion biopsy may be avoided. A lesion **>1.5 cm** or definite extraprostatic extension scores PI-RADS 5.

Feature	Peripheral zone	Transition zone
Dominant sequence	**DWI**	**T2WI**
Dominant score 3	Positive DCE upgrades to 4	DWI 5 upgrades to 4
v2.1 (2019)	—	T2WI 2 with DWI ≥4 becomes 3

KEY POINT

DCE never creates or downgrades PI-RADS 4–5 and never upgrades a TZ lesion. mpMRI misses a minority of significant cancers (mostly 3+4), so systematic cores remain relevant.

- Targeting is cognitive, software fusion or in-bore. Fusion-targeted cores are ≈3× likelier than systematic ones to find intermediate/high-risk disease.
- Take **2–4 cores per target** (AUA/SUO 2023: 2–3, with or without systematic cores) whenever biopsy is indicated and a PI-RADS ≥3 lesion exists.

3 · Systematic templates and technique

- **Extended 12-core** biopsy is the AUA standard, replacing sextant. Going from 6 to 12 cores raises detection, but 18–21 do not: **geometry, not core number**.
- Add TZ/anterior cores for repeat biopsy or gland >50 mL, and seminal vesicle cores for a palpable abnormality.
- Use an 18-gauge gun, and never reposition with the needle on the rectal wall.

4 · Transrectal vs transperineal (TP)

- TP passes clean perineal skin, causes no rectal bleeding and reaches apex and anterior gland (anterior-only cancers 6–20%).
- European Association of Urology 2023: TP is **preferred** (strong), with non-inferior detection. Give skin-flora cover or none; omitting antibiotics did not change the very low sepsis rate.
- Access is by brachytherapy grid, freehand guide (PrecisionPoint) or two 14-gauge cannulas.

5 · Antibiotic prophylaxis

- Every transrectal patient gets prophylaxis. AUA 2019: fluoroquinolone; 1st/2nd-generation cephalosporin with or without aminoglycoside; or 3rd-generation cephalosporin. Give ≥1 h before, for **≤24 h**.
- Single doses: levofloxacin 500 mg, gentamicin 5 mg/kg, amikacin 15 mg/kg, ceftriaxone 1–2 g. Aztreonam replaces aminoglycoside in renal impairment.
- The FDA 2016 fluoroquinolone warning covers tendon rupture, neuropathy, CNS effects and myasthenia.
- Infection risk factors are antibiotics within **6 months** (most consistent), diabetes, immunosuppression, prior UTI, travel and rectal resistant E. coli. These men need augmented or swab-targeted prophylaxis, or TP.
- Endocarditis risk needs IV ampicillin (vancomycin if allergic) and gentamicin. An enema has no proven infection benefit.

6 · Anticoagulation and anaesthesia

Contraindications are coagulopathy, severe immunosuppression, and acute urinary tract infection (UTI) or prostatitis.

Agent	Action
Low-dose aspirin	Continue
Warfarin, clopidogrel	Stop 5–7 days; biopsy when INR <1.5
Direct oral anticoagulants (DOACs)	Stop 2–5 days
Mechanical valve	Bridge with heparin

- Transrectal: periprostatic block, 1–2% lidocaine 5 mL each side at the seminal vesicle–prostate junction (beats intrarectal gel).
- Transperineal: local to the skin, then 5 mL into the **periapical triangle** and 3 mL along the capsule.

7 · Complications and management

Complication	Rate	Management
Haematuria	23–63%	Clot retention (0.7%): catheter, washout
Haematospermia	10–50%	Lasts ≈1 month
Rectal bleeding	2–22%	Ladder (below)
Infection / UTI	0.1–7% / 2–6%	Oral or IV antibiotics
Sepsis	**0.3–3.1%**	Carbapenem until cultures

Complication	Rate	Management
Vasovagal	1.4–5.3%	Trendelenburg, fluids
Retention	0.2–0.4%	Temporary catheter

ALGORITHM

1. **Post-biopsy fever**: assess for sepsis and take cultures.
2. Give **no empirical fluoroquinolone**, as E. coli is resistant. Start a carbapenem (or amikacin/3rd-generation cephalosporin by sensitivity) and resuscitate.

ALGORITHM

1. **Persistent rectal bleeding**: apply pressure.
2. If it continues, tamponade (sheath or Foley balloon). If still bleeding: endoscopy with adrenaline/sclerosant, then angioembolisation, then transrectal suture.

KEY POINT

Bleeding is the commonest complication overall; infection leads serious ones and admissions (1.0% rising to 4.1% in the resistance era). This explains universal prophylaxis and the move to the perineum.

8 · Repeat and saturation biopsy

P2 Saturation biopsy in CA prostate — >12 cores, template mapping, indications, no gain at initial biopsy

Not in the paper answer:

- Micro-ultrasound (29 MHz) approaches mpMRI without contrast (emerging).

ALGORITHM

1. Cancer yield on successive biopsies in ERSPC: **22% → 10% → 5% → 4%**.

- Take a third or fourth biopsy only for high suspicion, using mpMRI with targeting, ConfirmMDx, anterior/TZ cores and the TP route.

9 · Landmark trials

Trial	Design	Result	Take-home
PROMIS (2017)	576: mpMRI, TRUS, 5-mm template mapping (reference)	Sensitivity 93% vs 48%; negative predictive value 89% vs 74%; 27% avoid biopsy	mpMRI better triage
PRECISION (2018)	500 biopsy-naive: MRI with targeted biopsy vs 12-core TRUS	Significant cancer 38% vs 26%; 28% avoided biopsy	MRI pathway superior
MRI-FIRST (2019)	Paired systematic and targeted	Each alone missed cancers	Combine for maximal detection
PREVENT (2024)	TP without antibiotics vs transrectal targeted prophylaxis	Infection not significantly different; detection comparable	Both acceptable

10 · MCQ one-liners

MCQ ONE-LINERS

1. Absent seminal vesicle: ipsilateral renal agenesis in 79%.

2. Commonest false-positive hypoechoic lesion: BPH foci and prostatitis.
3. Müllerian cysts are midline; wolffian cysts are lateral.
4. Significant cancer in PROMIS: Grade Group ≥3 or core length ≥6 mm.

156 Pathology of Prostatic Neoplasia

1 · Precursor and intraductal lesions

P2 PIN — definition, histology, precursor evidence, LGPIN not reported, isolated HGPIN needs no repeat biopsy, cribriform PIN reclassified

Not in the paper answer:

- ~**20%** of high-grade prostatic intraepithelial neoplasia (HGPIN) carries the **TMPRSS2:ERG** fusion.
- **Intraductal carcinoma of the prostate (IDC-P)** is dense cribriform atypia with **basal cells retained**, usually **retrograde spread of high-grade invasive cancer**. It independently predicts early biochemical recurrence (BCR), and Grade Group (GG) 1 with IDC-P is upgraded at radical prostatectomy (RP) in **>80%**.

Lesion	Defining feature	Action on isolated finding
Low-grade PIN (LGPIN)	Mild atypia; irreproducible	Not reported
HGPIN	Benign architecture, atypical cells; non-cribriform	No repeat biopsy
PINATYP	HGPIN with adjacent small atypical glands	Manage as atypical
Atypical intraductal proliferation (AIP)	Atypia beyond HGPIN, below IDC-P	Consider re-biopsy (if benign or GG1 elsewhere)
IDC-P	Dense cribriform, basal cells retained	Definitive therapy (or re-biopsy); **excludes active surveillance (AS)**

KEY POINT

HGPIN is a weak risk marker; IDC-P usually marks high-grade cancer already present.
Traps: re-biopsy for HGPIN; surveillance for GG1 with IDC-P.

2 · Adenocarcinoma: location, spread, staging

- **Zones (McNeal)**: the peripheral zone (PZ) holds ~70% of cancers, including **85%** of T1c; they are posterolateral and palpable.
- The transition zone (TZ) holds ~20% of cancers, which are anterior, impalpable, under-sampled and lower grade. The central zone holds ~5%, rare but aggressive.
- Tumours are **multifocal in >85%**. With no true capsule, say "**extraprostatic extension**" (EPE); tumour exits via **perineural spaces**, posterolaterally.
- **Seminal vesicle invasion (SVI)** means tumour in the SV muscular wall, usually reached from the base through periseminal tissue.
- EPE is uncommon below **0.5 cm³**; nodes and SVI are uncommon below **4 cm³**.

Stage	Definition
T1a / T1b	Incidental at TUR of prostate (TURP)/enucleation; ≤5% / >5% of tissue
T1c	Non-palpable, found on needle biopsy
pT2	Organ-confined at RP
pT2+ (pT2x)	Positive margin at an intraprostatic incision or ill-defined boundary (apex)

Stage	Definition
pT3a	EPE; **includes microscopic bladder neck invasion**
pT3b	SVI
pT4	Other adjacent structures (rectum, levators, external sphincter, pelvic wall)

Trap: microscopic bladder neck invasion is pT3a; T1a/b depend on the 5% cut-off, not grade.

3 · Grading: Gleason and grade groups

P2 Gleason's score and its significance — history, scoring rules, significance, limitations

Not in the paper answer:

- **Biopsy score = commonest plus highest pattern**. The lowest score in practice is **6**, which has **no nodal metastatic capacity**.

Pattern	Morphology			
3	Discrete well-formed glands			
4	**Cribriform (incl. glomeruloid)**, poorly formed or fused glands			
5	Sheets, cords, single cells, solid nests, **comedonecrosis**			
	GG (International Society of Urological Pathology (ISUP); World Health Organization 2016)	Gleason	Nodes at RP	5-yr BCR-free
1		3+3	0%	**96%**
2		3+4	0.6%	**88%**
3		4+3	4.3%	**63%**
4		4+4, 3+5, 5+3	—	**48%**
5		9–10	—	**26%**

KEY POINT

The steep drop is between GG2 and GG3: never lump them as "7". 3+5 and 5+3 are GG4; GG5 starts at 9. Pure GG1 cannot metastasise, so calling it "cancer" is debated.

- Report **% pattern 4** for GG2–3 (ISUP 2014). GG2 with limited pattern 4 may suit AS.
- **Tertiary pattern 5** (<5% at RP) raises BCR risk but **stays in its GG**, because nodal risk is not raised. On biopsy any higher pattern counts as secondary, however small.

Issue	ISUP 2019	Genitourinary Pathology Society (GUPS) 2019
% pattern 4 in GG2–3; cribriform	Report	Report
IDC-P with invasive cancer	**Incorporated into grade**	**Not graded**; comment
Targeted cores of one lesion	**Aggregate grade** (plus per-core)	Each core; aggregate optional

KEY POINT

Cribriform pattern and IDC-P mark aggressive disease and argue against AS even in GG2, so they must be reported. Aggregate grading of a targeted lesion avoids under- or over-grading.

Upgrading setting	Risk
Biopsy GG1, then RP	~⅓ upgraded
Biopsy GG2, then RP	~20% become GG3
GG1 with negative mpMRI (PI-RADS 1–2)	Still ~20% at 6 yr

Reclassification within 3 yr usually means missed disease rather than progression.

4 · Needle biopsy assessment

- Use **separate jars per site**. Pitfalls: **at the base, SV/central zone mimic HGPIN; at the apex, Cowper gland mimics cancer**.
- Report the **number of positive cores plus one other extent measure**, and report **perineural invasion** on biopsy (it raises EPE risk).
- Basal cell markers, **high-molecular-weight cytokeratin (HMWCK) and p63**, stain benign glands, so their **absence supports cancer**. **AMACR** is an adjunct only.
- **Adenosis**, in the TZ, is the chief mimic of low-grade cancer and carries **no increased cancer risk**.
- **Atypical small acinar proliferation (ASAP)**: cancer follows in ~**40%**. Get **expert review before re-biopsy**, then re-biopsy the atypical site.

> **KEY POINT**
> Sampling error is one-directional: a bad biopsy is believed, a good one is provisional, and an atypical one is re-examined before it is repeated.

5 · TUR and RP specimens

- At TUR, **grade each major nodule**. Nodes are uncommon with Gleason <8.
- **EPE**: focal vs nonfocal extent predicts progression. **SVI** carries **65%** 5-yr progression.
- **Margins** are positive in ~15% (robotic), but only ~50% progress. Report **extent (focal vs nonfocal, 3 mm) and grade at the margin**.
- Independent predictors of BCR are grade, EPE extent, margins, SVI and nodes. **Tumour volume is not independent**.

6 · Treatment effect

Treatment	Histological effect	Gleason score?
Hormone therapy	Grade looks artefactually higher	**No** where effect is present
Finasteride	No consistent change	Yes
Radiotherapy	Radiation atypia (up to 72 months)	Only if no treatment effect
Cryotherapy / high-intensity focused ultrasound (HIFU)	Infarction; residual cancer unaltered	**Yes**

After radiotherapy, cancer with treatment effect has a prognosis close to no cancer.

7 · Subtypes and other tumours

Subtype	Grading	Behaviour
Ductal (pure 0.4–0.8%; mixed ~5%)	**Pattern 4** (PIN-like 3; comedonecrosis 5)	Exophytic at the verumontanum; PSA may be normal; behaves like Gleason 8; **visceral recurrence at low PSA**
Mucinous	As usual	As acinar
Neuroendocrine differentiation (~half of acinar)	As usual	Not prognostic
Small cell (~50% mixed)	**Not graded**	Survival <1 yr
Squamous	—	**Osteolytic** metastases; hormone-unresponsive
Sarcomatoid	—	Dismal

- **Urothelial carcinoma in the prostate**: suburethral invasion is pT1 and stromal pT2, but **transmural spread from the bladder is pT4**. Urothelial cancer stains **GATA3**; prostate stains PSA and NKX3.1.
- Sarcomas are **rhabdomyosarcoma in children and leiomyosarcoma in adults**.

157 Diagnosis and Staging of Prostate Cancer

1 · Clinical presentation

- Early cancer is **asymptomatic**; symptoms imply advanced disease.
- **Locally advanced** disease causes voiding or ureteric obstruction and haematuria; **metastatic** disease causes bone pain, fracture, leg oedema and cord compression.

2 · Screening in brief

P2 PSA screening for prostate cancer: evidence, problems and solutions — ERSPC vs PLCO, NNI 570 / NND 18, USPSTF D to C, AUA, NCCN, overdiagnosis

3 · DRE and PSA

- **Digital rectal examination (DRE)** has low screening value at PSA <3, but an abnormal DRE raises the risk of Grade Group (GG) 4–5 at every PSA level.
- **Prostate-specific antigen (PSA)** is organ-specific, not cancer-specific, and rises ≈4% per mL of volume.
- PSA is lowered by hypogonadism, obesity, statins and 5α-reductase inhibitors (5-ARIs). On a 5-ARI, **double PSA** after ≥12 months, or treat the nadir as the new baseline.
- Risk is **continuous**: the Prostate Cancer Prevention Trial (PCPT) found cancer in 15% of men with PSA ≤4.
- Thresholds are 4 or 2.5–3 (ERSPC used 3). Repeat a raised PSA before biopsy, and give no empirical antibiotics without urinary symptoms.

PSA (ng/mL)	Yield, normal DRE	Yield, suspicious DRE	High-grade (normal / suspicious)
0–1	8.8%	—	0.9%
4–10	17–32%	45–51%	4.1% / 11.7%
>10	43–65%	70–90%	19.4% / 50.5%

- Baseline PSA in the 40s predicts metastasis and death decades later (Malmö).
- Derivatives: PSA density ≥0.15; velocity >0.75 ng/mL/yr; free PSA ≤25% (at PSA 4–10). See Ch 154, 155

4 · Staging: TNM (AJCC 8th, 2017)

Pathological stage is more accurate and prognostic than clinical stage.

Category	Definition
T1a / T1b	Incidental at transurethral resection (TUR): ≤5% / >5% of tissue
T1c	Impalpable, found on needle biopsy
T2a / T2b / T2c	Palpable, confined: ≤½ of one side / >½ of one side / both sides
T3a / T3b	Extraprostatic extension (EPE) / seminal vesicle invasion (SVI)
T4	Fixed, or invading sphincter, rectum, bladder, levators or pelvic wall
N1	Regional (pelvic) nodes
M1a / M1b / M1c	Non-regional nodes / bone / other sites with or without bone

AJCC prognostic groups add PSA and GG.

KEY POINT

T1 is defined by how the cancer was **found**, T2 by what is **felt**. Traps: SVI is T3b; non-regional nodes are M1a; a biopsy-found impalpable cancer is T1c even if bilateral; clinical T is set by DRE, not imaging.

5 · Predicting extent and risk

- **PSA** cannot stage an individual. Organ-confined disease is ≈80% at PSA <4 and <50% above 10; nodes are positive in **75% at >50**.
- **Biopsy grade** is the most important biopsy finding, and tertiary pattern 5 on biopsy is recorded as the secondary pattern. See Ch 156

D'Amico	Stage	PSA	Gleason	10-yr disease-free after RP
Low (all criteria)	T1–T2a	≤10	≤6	83%
Intermediate (any one)	T2b	>10–20	7	46%
High (any one)	T2c	>20	8–10	29%

National Comprehensive Cancer Network (NCCN) group	Criteria
Very low	T1c, GG1, PSA <10, PSA density <0.15, <3 positive cores, ≤50% of any core
Low	T1–T2a, GG1, PSA <10
Favourable intermediate	One intermediate factor (T2b–c, GG2–3, PSA 10–20), GG1–2, <50% cores positive
Unfavourable intermediate	≥2 intermediate factors, or GG3, or ≥50% cores positive
High	T3a, or GG4–5, or PSA >20
Very high	T3b–T4, primary pattern 5, >4 cores GG4–5, or ≥2 high-risk features

- CAPRA and **nomograms** predict better than groups for individuals; nodal nomograms decide pelvic lymph node dissection (PLND). Tissue genomic tests refine risk but await prospective validation.

6 · Staging imaging and nodes

Risk	Imaging
Low / intermediate, asymptomatic	No routine computed tomography (CT) or bone scan
High	Bone scan with CT or mpMRI; consider PSMA PET

- **Multiparametric MRI** is used for local staging (EPE, SVI), treatment planning and surveillance selection.
- **PSMA PET** (68Ga-PSMA-11, piflufolastat F-18) is FDA-approved for high-risk staging and is more sensitive and specific for nodes and metastases.
- **PLND** has no consistent oncological benefit; its value is **staging**. Do it when the nomogram nodal risk exceeds the guideline threshold.

KEY POINT

Imaging yield follows pre-test probability: in low-risk men most positive scans are false positives, leading to more tests, anxiety and wrongly denied cure.

7 · Landmark trials

Trial	Design	Result	Take-home
ERSPC	European screening randomised trial, from 1993	**20% relative reduction** in cancer death at 16 y	Saves lives, at the cost of overdiagnosis
PLCO	US screening randomised trial, from 1993	No mortality difference	Contaminated control arm
PCPT	Placebo arm biopsied regardless of PSA	15% cancer at PSA ≤4	No safe PSA level
proPSMA (2020)	PSMA PET-CT vs CT and bone scan, high-risk	Accuracy ≈92% vs 65%; less radiation	PSMA PET is the better first-line staging test

8 · MCQ one-liners

MCQ ONE-LINERS

1. Finasteride 1 mg lowers PSA as much as 5 mg.

158 Active Management Strategies for Localised Prostate Cancer

1 · The modern question: timing as well as modality

- In 15-year randomised data on favourable risk, surveillance gave **more metastasis but no difference in cancer mortality**.
- **Spectrum**: a man likely to die of competing risks gets **watchful waiting (WW)**; good longevity with unclear benefit means **active surveillance (AS)**; a man who needs therapy gets radical prostatectomy (RP), external-beam radiation therapy (EBRT), brachytherapy, or whole/partial-gland ablation.

> **KEY POINT**
> Cancer mortality is always weighed against competing mortality. A competing risk argues against treatment only when it will act before the cancer does.

2 · Host evaluation

- NCCN estimates life expectancy from Social Security Administration or WHO tables.
- In the Prostate Cancer Outcomes Study (PCOS), 10-year prostate cancer mortality was **3–18%** by risk, but other-cause mortality was **≥33%** with any comorbidity.
- The blunt **"10-year life expectancy" rule** undertreats older and overtreats younger men, so individualise.
- The **Charlson index** (1-year inpatient mortality) is reweighted as **PCCI** for 10-year mortality; NNT rises steeply with Charlson score.
- The automated **Care Assessment Needs (CAN) score** (Veterans Health Administration) best predicts 5-year survival MCQ.

Baseline function (IPSS, UCLA Prostate Cancer Index)

Baseline state	Favours	Reason
Obstructive lower urinary tract symptoms (LUTS), large gland	Surgery	RP relieves obstruction; brachytherapy increases retention
Severe irritative LUTS	Surgery	Worst after radiation
Gland >60 cm^3	Surgery (or neoadjuvant androgen-deprivation therapy, ADT)	Pubic arch interference; if >80 g, ADT (shrinks 25–40%) before radiotherapy (RT)
Chronic retention / neurogenic bladder	Surgery	RT causes bladder fibrosis; if RT is chosen, switch to intermittent self-catheterisation
Prior transurethral resection (TUR) defect	Surgery	Brachytherapy dosimetry is hard and incontinence increases. Do TURP before RT and delay EBRT **6–8 weeks**
Inflammatory bowel disease (IBD), bowel dysfunction, prior rectal surgery	Surgery	RT bowel toxicity

Baseline state	Favours	Reason
Poor sexual function	—	Recovery depends on age, baseline erection, PDE5 inhibitor use and vascular risk; poor baseline function means non-nerve-sparing surgery is used more.

Modality	Absolute contraindication	Relative contraindication
Radiotherapy	Previous radical pelvic irradiation	Bowel in field (prior surgery, hernia); uncontrolled diabetes; IBD; bowel dysfunction
Surgery	—	Previous pelvic RT; chronic steroids; severe cardiac/pulmonary/haematological comorbidity

3 · Risk stratification

NCCN group	Criteria	Staging imaging
Very low	**All**: T1c; grade group (GG) 1; prostate-specific antigen (PSA) <10; <3 positive cores, ≤50% each; PSA density < **0.15**	Not indicated
Low	**All**: T1–T2a; GG1; PSA <10	Not indicated
Favourable intermediate	Intermediate-risk factors (IRFs): T2b–c, GG2, PSA 10–20. **All**: 1 IRF; GG1–2; <50% cores positive	
Unfavourable intermediate	**Any**: 2–3 IRFs; GG3; ≥50% cores positive	Bone scan and CT/MRI, plus chest imaging; prostate-specific membrane antigen (PSMA) PET may replace these (NCCN 2023)
High	**Any**: T3a; GG4–5; PSA >20	
Very high	**Any**: T3b–T4; primary pattern 5; >4 cores GG4–5	

KEY POINT

Favourable groups need ALL criteria; adverse groups ANY one. PSA density appears only in very low risk; GG3 alone makes intermediate unfavourable. AUA/ASTRO/SUO use very low/low/intermediate/high without substratification.

- **Epstein criteria** (insignificant cancer): PSA density <0.15, Gleason ≤6, <3 positive cores, ≤50% of any core.
- Nomograms: **Partin** predicts pathological stage (accurate 72.4%); **Kattan** predicts relapse after RP; with **UCSF-CAPRA**, risk jumps at score ≥6.
- **proPSMA** (302 high-risk men): PSMA PET had accuracy **92% vs 65%** and sensitivity 85% vs 38% over conventional imaging, but a negative scan cannot exclude small nodes.
- Genomic tests are not routine (AUA 2022); use them only where they change an AS decision.

4 · Observation vs active surveillance

P2 Expectant treatment of CA prostate — AS (curative, deferred) vs WW (palliative): candidate, monitoring, triggers

Not in the paper answer:

- AS is **preferred for very low and most low risk** (AUA, ASTRO, SUO, NCCN); selected low-volume GG2 is acceptable.
- Limitations are **25–45%** occult high grade on systematic biopsy (mostly GG2), declining adherence and no agreed triggers. PSA density >0.15 predicts reclassification.

5 · Comparative oncological evidence

- Population studies favour RP (Swedish subdistribution HR ≈1.8; PCOS disease-specific HR 0.35, curves apart by 3 years), but this is largely **selection bias**.

6 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	RP vs WW; 695; pre-PSA; ~90% palpable	23.2 yr: cancer deaths 63 vs 99, relative risk (RR) **0.56**; absolute 11%; NNT **8**; most benefit <65 (RR 0.45), intermediate risk	Benefit in younger men with intermediate risk
PIVOT	RP vs observation; 731 (2000 planned); PSA era; older, comorbid	10 yr: no difference; benefit if PSA >10. 19.5 yr: all-cause ARR 5.5% and cancer 4.0% (both not significant); significant only in **intermediate risk** (ARR 14.5%). Bone metastases 4.7% vs 10.6%	No benefit in low risk; underpowered
ProtecT	1643 screen-detected, 50–69; monitoring vs RP vs RT with 3–6 mo ADT; 77% low risk	15-yr cancer mortality **2.7% vs 2.2% vs 2.9%** (P = 0.53); more metastasis with monitoring; 60% of monitored treated by 15 yr	AS safe in favourable risk; RP equals RT

KEY POINT

Competing mortality and low-risk enrolment dilute trial benefit, which appears only in younger intermediate-risk men with long follow-up.

7 · Functional outcomes compared

Cancer control is similar, so side effects usually decide.

Domain (vs AS; Chen, JAMA 2017)	RP	EBRT	Brachytherapy
Sexual	**Worst**; persists at 24 mo	Mild early decline; near AS by 24 mo	Mild early decline
Incontinence	**Worst**; persists at 24 mo	No change	No change
Obstructive/irritative	Worse at 3 mo, then **better than AS**	Worse at 3 mo; resolves by 12	**Worst acute**; resolves by 24 mo
Bowel	No change	**Worse**; persists at 24 mo	No change

- **PCOS** found differences at 2 and 5 years but **none at 15 years**. **ProtecT** found rising bloody stools after RT, with no difference in anxiety, depression or quality of life.
- In **CEASAR**, the erectile disadvantage of RP was confined to the top quartile of baseline function. In open vs robotic RP, the surgeon matters more than the approach.

KEY POINT

Surgery front-loads urinary and sexual harm that partly recovers; radiation spares early function but adds bowel toxicity and slow decline. Curves converge by 15 years.

8 · Shared decision-making

- No treatment is clearly superior, so **patient preference carries unusual weight** (AUA, ASTRO, SUO).
- The **urologist's recommendation is the most influential factor**, so beware your own bias. **Longevity assessment may be the most important part of counselling**.

9 · Stepwise decision algorithm

ALGORITHM

1. **Treat or not?** Weigh disease and host factors. If longevity is short, then WW; if very low/low or selected favourable intermediate, then AS.
2. **Partial or whole gland?** Weigh focality, size, grade, potency, LUTS. If limited-focality intermediate risk, then focal ablation, ideally in a trial.
3. **Surgery, radiation or whole-gland ablation?** Compare outcomes, toxicities, baseline function and risk of secondary treatment.
4. **Delivery:** RP approach and node dissection, or EBRT vs brachytherapy, fractionation and ADT.

Risk / longevity	Usual recommendation
Limited longevity, low lethal potential	WW; ADT at symptoms
Very low	AS (best option)
Low	AS preferred for most; all therapies are options
Favourable intermediate	RP or RT; AS for select (low-volume GG2); focal therapy in trials
Unfavourable intermediate	RP, or RT with ADT; AS strongly discouraged
High / very high	RP, or RT with ADT; consider even with competing risks if the cancer threatens within 10 yr; no focal therapy

- **Surgery:** early incontinence and erectile dysfunction (ED) are the norm, so quote your own outcomes; LUTS may improve.
- **Radiation:** it spares short-term function but causes yearly decline; secondary rectal/bladder cancer at 15–20 years is controversial.
- **Whole-gland ablation:** long-term data are scant and fibrosis hampers salvage. Cryoablation almost always causes ED; HIFU preserves function better, but retreatment is **29–38%**.
- EBRT, brachytherapy, SBRT and protons give comparable 10-year cancer control.

159 Active Surveillance of Prostate Cancer

1 · Rationale: overdiagnosis

- **Latent reservoir**: autopsy cancer prevalence roughly equals age as a percentage. PSA testing and biopsy detect cancer in 15–25%, yet lifetime risk of cancer death is only **3%**; overdiagnosis is 87–94%.
- Active surveillance (AS) rests on **two premises**: progression takes a long time, and cure is still possible at the first sign of higher risk.

2 · AS versus watchful waiting

P2 Expectant treatment of CA prostate — AS curative/deferred vs WW palliative; candidates, monitoring, switch at life expectancy <5–10 yr

Not in the paper answer:

- The switch to watchful waiting (WW) is typically near age 80.
- **Pitfall**: elderly men with unrecognised hypogonadism may progress (neuroendocrine dedifferentiation) without a PSA rise, so measure testosterone and get interval imaging.

> **KEY POINT**
> AS defers curative intent; WW abandons it. Life expectancy, not grade, chooses between them.

3 · Biology of grade group 1

- **Pattern 3 is molecularly near-normal.** Patterns 4–5 show EGFR, HER2 and BCL2 changes and **PTEN deletion in 70–90%**; **MYC amplification with PTEN deletion** drives metastasis.
- **Metastatic potential of pure grade group (GG) 1 is effectively zero**, though occasional extraprostatic extension occurs.
- True grade progression is ~**1–2% per year**, mostly in high-volume Gleason 6, so follow-up must be long term.
- Deaths in low-risk series come from **occult higher grade** missed at biopsy, present in ~**30%** after systematic biopsy (85% of it 3+4).

Hazard	Misattribution of grade	True progression
Frequency	**Commonest (25–30%)**	1–2.3%/year
Usual result	Mostly GG2	Many 3+4 stay indolent
Remedy	MRI, targeted biopsy, predictors	Long-term surveillance

> **KEY POINT**
> The main hazard of AS is **sampling error at entry**, not biology over time. First job: get the entry grade right.

4 · Patient selection

Most GG1 should be on AS (AUA, EAU, ASCO, NICE and Cancer Care Ontario agree).

	Very low risk (Epstein)	Low risk
Definition	GG1; **≤2 positive cores; no core >50%; PSA density <0.15**	Any GG1, PSA <10
Progression to treatment	Lower	Higher
Metastasis	Exceptionally low	Exceptionally low

High-volume GG1 is still a candidate, but check it with multiparametric MRI (mpMRI) and targeted biopsy, re-biopsy or a genomic assay.

- **Toronto** is inclusive (all Gleason 6 and selected intermediate), with 15-yr cancer mortality of **5%** actuarial; **Johns Hopkins** is restrictive (Epstein very low), with **0.5%**. So **inclusion criteria set mortality, while intervention stays constant (~25% at 5 yr)**.
- In Toronto, baseline Gleason 7 had ≥20% metastasis at 15 yr (HR 3.75). In PRIAS (n=5302), cancer mortality was <1% at 10 yr, and cardiovascular disease was the commonest cause of death.
- **Favourable intermediate**: offer AS selectively for small-volume GG2. Only NICE lists AS and radical treatment as equal for intermediate risk.
- Exclude GG ≥3; intraductal or cribriform histology raises concern.
- **Black men** are AS candidates, perhaps with closer surveillance. **Family history** rarely justifies treatment, except metastatic disease <60 yr in a first-degree relative, which calls for germline testing.
- **BRCA2** carries the highest upgrading risk; consider definitive therapy even for low grade, or AS only with very close follow-up.
- Risk factors for higher grade are **PSA density >0.15**, Black race and biopsy volume; restricted diffusion on MRI is a powerful predictor.

5 · Monitoring protocol

ALGORITHM

1. At diagnosis, obtain mpMRI.
2. Do a **confirmation biopsy** (systematic + targeted) within 1–3 yr.
3. Check PSA 6-monthly, DRE every 1–2 yr and MRI ≤ every 18–24 months.
4. Repeat biopsy every 3–5 yr while stable.
5. If a trigger appears, then give definitive therapy; if life expectancy falls, then switch to WW.

- Confirmation biopsy may be omitted if the first biopsy was MRI-targeted. **Compliance is paramount**: compliers had better cancer mortality and metastasis-free survival.

Guideline	Intermediate risk	Schedule / MRI
Cancer Care Ontario	Treat; AS if selected	PSA 3–6-monthly; DRE yearly; biopsy within 6–12 mo then 3–5-yearly; MRI if discordant
AUA/ASTRO 2022	AS if selected	PSA 6–12-monthly; DRE 1–2-yearly; biopsy 1–4-yearly; MRI before initial or confirmation biopsy
NICE 2019	Radical treatment or AS	PSA 3–4-monthly in year 1, then 6-monthly; **MRI at enrolment**

Prompts definitive therapy	Prompts evaluation only
GG ≥3 (unless life expectancy significantly shortened)	More Gleason 6 volume: get MRI or a biomarker
Traditionally any Gleason ≥7 (now relaxed)	Small-volume GG2: may continue AS
	PSA rise alone: repeat MRI or biopsy

KEY POINT

Grade triggers treatment; volume and PSA trigger investigation. 10–15% leave AS by choice (anxiety) without a trigger.

6 · MRI in AS

- **At entry the evidence is strong**: mpMRI before confirmation biopsy, with **systematic and targeted cores**, upgrades about ⅓. NICE recommends it; NCCN lists it as an option.
- **Serial MRI alone is weak** (NPV ~0.8–0.9 for GG ≥2), so it must never be the only way to exclude progression. MRI-invisible cancers behave better.

7 · Biomarkers and genomics

- Decipher predicts upgrading in GG1, not GG2; Oncotype DX evidence is mixed and Prolaris has none specific to AS.
- **Bayesian problem**: with a 3% prior metastasis risk, a 90%-specific test labels 10% "significant", so test only higher-risk candidates (high volume, high PSA density, GG2).
- Use the tests to set surveillance intensity, **never as the sole basis for AS**; a favourable score does not override adverse features.

8 · Lifestyle, drugs and psychology

- **Stop smoking**, exercise vigorously (ERASE: lowered PSA), keep a healthy weight and eat a vegetable-based diet.
- Only vitamin D 1000–2000 IU/day is reasonable. **Selenium and vitamin E are contraindicated** (SELECT).
- 5α-reductase inhibitors are not used to prevent progression (REDEEM); metformin, apalutamide and enzalutamide are investigational.

9 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	695, pre-PSA, palpable; RP vs WW; 23 yr	Cancer death 19.6% vs 31.3% (relative risk 0.55)	Benefit at age <65 and GG ≥3; none in GG1–2
PIVOT	731 US veterans; RP vs observation; 18.6 yr	All-cause death 68% vs 73% (not significant); low-risk cancer death 2.7%	Modest benefit in intermediate risk
ProtecT	1643 PSA-detected; monitoring vs RP vs radiotherapy; 15 yr	Cancer mortality <3%, no difference; **metastases 9.4% vs 4.5% vs 5.0%**; 61% of monitoring arm treated	Active monitoring is not AS (no scheduled biopsy, >⅓ intermediate/high)

Trial	Design	Result	Take-home
ASIST	GG1 on AS; MRI-targeted + systematic vs systematic confirmation biopsy	Upgrading 33% vs 27% (not significant); ~50% less AS failure at 2 yr	MRI benefit shows over follow-up
REDEEM	Dutasteride vs placebo on AS	38% less "progression" (less GG1 volume), no upgrading difference	Cytoreduction, not disease modification

160 Open Radical Prostatectomy

1 · Principles

- Radical prostatectomy (RP) alone has been shown in a randomised controlled trial (RCT) to reduce metastasis and cancer death (**SPCG-4**). Most RPs are now robotic, without proven benefit.
- **Goals, in order**: cancer control, then avoiding technical complications, then continence, then potency.

> **KEY POINT**
> The order of goals settles every intraoperative trade-off: when nerve sparing and a clear margin conflict, the margin wins.

2 · Surgical anatomy

P1 P4 Anatomy of prostate and neurovascular bundles; Prostate anatomy, newer surgical concepts and NVB preservation — fascial layers table, NVB of Walsh, inferior vesical urethral/capsular branches, accessory pudendals, Santorini plexus, nerve-sparing planes

Not in the paper answer:

- Inferior vesical urethral branches enter at the bladder neck (**5 and 7 o'clock** pedicles); capsular branches mark the cavernous nerves.
- The pelvic plexus combines parasympathetic **S2–S4** and hypogastric sympathetic fibres. It lies beside the rectum, **5–11 cm** from the anal verge, centred on the seminal vesicle tip.
- The **striated sphincter** is a vertical tubular sheath, not a flat diaphragm, so apical dissection decides continence; it and the levator ani are both mainly **pudendal**.

> **KEY POINT**
> Nerves and vessels run between levator and prostatic fascia. Leaving prostatic fascia on the gland and Denonvilliers fascia on its back lets nerve sparing and clear margins coexist.

3 · Selection and preoperative care

- **Harder, not contraindications**: prior pelvic surgery or radiotherapy, transurethral resection of the prostate (TURP), hernia mesh, large gland.
- **Defer surgery 6 weeks after biopsy and 12 weeks after TURP**, so haematoma settles.
- Inguinal hernia affects ~15% before and up to 20% after RP, so examine standing and repair it preperitoneally at RP.
- Give cephalosporin prophylaxis; the **fibreoptic headlight** is the most useful instrument.

4 · Retropubic RP — operative steps

> **ALGORITHM**
> 1. Lower midline extraperitoneal incision; pelvic lymph node dissection (PLND).
> 2. Open endopelvic fascia; divide puboprostatic ligaments.
> 3. Control and divide the DVC, then the urethra and posterior sphincter.
> 4. Release the NVBs; divide pedicles, bladder neck and seminal vesicles.
> 5. Reconstruct the bladder neck and make the anastomosis.

- **PLND** is for intermediate/high risk, omitted for Grade Group 1–2 low-volume disease.
- **Endopelvic fascia**: open at the sidewall reflection, because too medial causes Santorini bleeding. Divide the **puboprostatic ligaments** but spare the **pubourethral** part, which anchors the sphincter.

DVC control

Feature	Stapler (preferred)	Suture ligation
Method	Vascular load; fire only if the catheter moves freely	3-0 figure-of-eight to pubic perichondrium, plus a back-bleeding stitch
Advantage	~200–400 mL less blood loss; less traction	Recreates puboprostatic support

- The **anterior apex is the commonest positive margin site**, so get absolute venous control before touching it.
- **Urethra**: transect close to the apex and place six 3-0 sutures through **mucosa and submucosa only**, because muscle bites delay continence.

Nerve sparing

- The **interfascial** plane (levator vs prostatic fascia) is correct; Walsh never uses intrafascial because of margin risk. Wide (extrafascial) excision is for induration, a fixed bundle or doubtful margin.
- Both NVBs are usually kept, because extracapsular extension rarely exceeds 1–2 mm. Decide in theatre: **if induration or a fixed bundle, excise that side**.
- **Technique**: release the prostate from the bundle, not the bundle from the prostate, starting **at the apex**. Use small clips and **never thermal energy**.
- **High anterior release** (before urethral division) means less traction and faster recovery.
- Avoid it if ≥2 **Tsuzuki criteria** are present: PSA >10, Gleason >6, >20% core involvement, >33% cores positive, perineural invasion, abnormal DRE.
- For **wide excision**, clamp **medial to lateral** on the rectum, since the reverse risks rectal injury.

Pedicles, bladder neck, reconstruction

- **Complete seminal vesicle excision is essential**; the pelvic plexus lies on their lateral surface.
- Close the bladder neck as a **tennis racquet** with mucosal eversion, and make a tension-free anastomosis; a peritoneal window lets haematoma decompress.

> **KEY POINT**
> Every step does one of three jobs: absolute venous control before the apex; the right fascia on the specimen; no traction or heat on the NVB.

5 · Perineal RP (brief)

- Perineal RP reflects the fasciae off the prostate and spares the DVC, so it bleeds less, but PLND needs a separate approach.

6 · Intraoperative complications

Complication	Setting	Management
Haemorrhage (commonest, venous)	Endopelvic fascia, DVC	**Divide DVC completely and oversew** (best measure)
Obturator nerve	PLND	Reanastomose with fine non-absorbable suture
Rectal injury	Small prostate; apical/posterior plane	Two-layer repair and omentum; colostomy if prior radiotherapy
Ureteric injury	Dissecting in trigone	Reimplantation

- Typical loss is 300–1000 mL; partial DVC division bleeds more.
- **Rectal repair**: finish the prostatectomy, dilate the anus, close in 2 layers and place omentum between rectum and anastomosis to prevent a rectourethral fistula.

7 · Postoperative care

- Discharge on day 1–2. Give ketorolac only if creatinine <1.2 mg/dL and there is no haematuria.
- Keep the **catheter ~1 week**, because day-4 removal causes frequent retention; never remove it with significant extravasation.

8 · Postoperative complications

- Mortality is **0.2%**. **Delayed haemorrhage** needs **early exploration and evacuation**, which shortens stay.
- **Thromboembolism** is the major cause of death. It is highest with PLND and peaks at **14–28 days**. Use compression and early ambulation; treat with low-molecular-weight heparin (LMWH).
- **Bladder neck contracture** appears at ~3 months. Treat with dilatation, then cold-knife incision at 3, 6 and 9 o'clock with self-catheterisation and triamcinolone; reconstruction with an artificial sphincter is rarely needed.

P2 Post-prostatectomy incontinence — rhabdosphincter dependence, evaluation, PFMT, sling vs AUS

Not in the paper answer:

- Continence means **0–1 pad/24 h**. Prevent incontinence with a preserved apical sphincter, a precise tension-free mucosal anastomosis and a small supple neck.
- **Erectile dysfunction (ED)**: recovery is predicted by **age <65, preoperative potency, no diabetes and bilateral nerve sparing**, and continues up to 4 years. PDE5 inhibitor rehabilitation helps; nightly is not better than on-demand.

> **KEY POINT**
> Most late complications trace to the pelvic haematoma: it distracts the anastomosis (contracture, incontinence) and inflames the NVBs. Hence early re-exploration, clips near the bundles and a peritoneal window.

9 · Outcomes

Time	Continent	Potent (Walsh)
3 months	80%	38%
12 months	93%	73%
18–24 months	97%	86%

- These are single-expert figures; elsewhere only a third regained baseline potency by 2 years. RP improves lower urinary tract symptoms for up to 15 years.

10 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	RCT, RP vs observation	Less metastasis and cancer death	RP cuts cancer death
Padma-Nathan	RCT, nightly sildenafil vs placebo	Better erections at 1 year	Rehabilitation works
REINVENT	RCT, vardenafil on-demand vs nightly	On-demand better	Daily dosing unproven

161 Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy

1 · Background

P1 Open vs laparoscopic vs robotic radical prostatectomy; Robotic radical prostatectomy — history, comparison table, contraindications, ports, steps, complication rates

Not in the paper answer:

- Avoid relative contraindications (including neoadjuvant hormones) early in the learning curve.

> **KEY POINT**
> The robot did not change the operation; it removed the suturing barrier. Surgical anatomy (dorsal venous complex [DVC], neurovascular bundle [NVB], fasciae, sphincter) is as in open surgery.

2 · Patient selection

- The only absolute contraindications are uncorrectable bleeding diathesis and unfitness for general anaesthesia.
- After extraperitoneal mesh repair, use the **transperitoneal** route.
- **Obesity**: place longer trocars more cephalad and lower pneumoperitoneum to 10 mm Hg. A **large gland or median lobe** takes longer and bleeds more, but continence is comparable; **prior TURP** brings the ureteric orifices close to the bladder neck.
- Multiparametric MRI (mpMRI) for extracapsular extension (ECE) has specificity **0.88** and sensitivity **0.61**; it guides nerve sparing.

3 · Preparation and anaesthesia

- Consent for **conversion to open**; **general anaesthesia is mandatory**.
- **Positioning**: tuck and pad arms; use **no fixed shoulder rests** (brachial plexus injury) and avoid exaggerated hip flexion (femoral neuropraxia).
- CO2 pneumoperitoneum causes vagal response, oliguria and hypercarbia, so raise ventilation and lower pressure. Restrict fluids to 1.5–2 L, mostly after the anastomosis.
- **Steep Trendelenburg** causes facial, laryngeal and corneal oedema and raises intraocular pressure, so protect the eyes and consider delayed extubation if the face is very swollen.

4 · Access

- **Extraperitoneal** access gives a smaller space, protects bowel and needs less tilt (suits obesity and lung disease), but absorbs more CO2. **Transperitoneal** gives a larger space. Neither has a consistent advantage.
- Transperitoneal **anterior** (commonest; Retzius first) differs from **retrovesical** (seminal vesicles first, useful with a median lobe).
- Place the camera port **15–17 cm** from the symphysis; insufflate to 12–15 mm Hg.

5 · Operative steps (transperitoneal anterior RALRP)

ALGORITHM

1. Drop the bladder (Retzius); open the endopelvic fascia; ligate the DVC.
2. Divide the bladder neck; free the seminal vesicles and vasa.
3. Develop the posterior (Denonvilliers) plane; control pedicles and release the NVBs.
4. At the apex, divide the DVC and urethra; inspect the specimen.
5. Do pelvic lymph node dissection (PLND); entrap the specimen.
6. Posterior reconstruction, anastomosis, extraction.

- Open the endopelvic fascia with **no cautery**. Pneumoperitoneum tamponades DVC bleeding, so it can be sutured, stapled, or divided late and oversewn.

Bladder neck — no tactile feedback

- Divide anteriorly in the midline, lift the catheter, inspect for median lobe and orifices, then divide the posterior neck angled **45°** downward.
- After TURP the balloon falls into the defect, so division drifts too distal; a balloon off the midline suggests a median lobe. When in doubt, **divide more proximally**.

Posterior plane, pedicles, nerve sparing

P1 P4 Anatomy of prostate and NVB; NVB preservation in RP — nerve-sparing planes, Denonvilliers excision for posterolateral ECE, injury mechanisms, apex = commonest margin

Not in the paper answer:

- **Athermal (veil) release**: incise levator fascia anteromedially and sweep the NVB posterolaterally off the prostatic fascia, then Hem-o-lok the pedicle and divide sharply. The **intrafascial** plane is accepted in selected men.
- Pedicle clips near the anastomosis can migrate into the bladder, causing haematuria, infection and stones.

Apex and reconstruction

- Preserve urethral length; a small urethral rim on the apex lowers margins without harming continence.
- Reconstruct the bladder neck (tennis racquet) only for a large gland, median lobe or prior TURP, and protect the orifices.
- The **Rocco stitch** joins the Denonvilliers remnant and posterior bladder to the posterior rhabdosphincter. Its continence benefit is inconsistent; its agreed benefit is a **tension-free anastomosis**.
- The **Van Velthoven anastomosis** runs two arms anteriorly from 6 o'clock.

6 · Continence-preserving variants

P4 Newer surgical concepts in CA prostate — Retzius-sparing: earlier continence, more margins; Hood: early data

- Retzius-sparing leaves the detrusor apron, puboprostatic ligaments and DVC untouched, on the idea that preserving anterior support beats reconstructing it.

7 · Pelvic lymph node dissection

P4 P1 PLND templates; Lymphocele — templates (extended ~63%), staging not survival; lymphocele drain, then fenestration

Not in the paper answer:

- **Indications**: may omit in low risk (nodes <5%). AUA 2018: unfavourable intermediate or high risk. EAU 2021: nomogram risk **>5%**. AUA/ASTRO 2022: risk-stratified by nomogram (Briganti and MSKCC nomograms predate PSMA PET).
- The standard template misses most drainage, so **if PLND is done, use the extended template**: proximal 2 cm above the common iliac bifurcation (ureter crossing), distal node of Cloquet, lateral **genitofemoral nerve**, medial bladder wall.
- **Clipping lymphatics** means fewer lymphoceles; node count is a poor quality surrogate.

PLND complications (~1 in 7)

- For the obturator nerve, remove a clip or repair a transection primarily. Minimise heat proximally, where the ureter is at risk.
- **Lymphocele is commonest** (**2–5%** significant). **Transperitoneal access is not protective**, because the bladder walls off the fossa; a **peritoneal interposition flap** reduces it. It classically presents with ipsilateral deep vein thrombosis (DVT).

8 · Postoperative care

- Use enhanced recovery, with same-day discharge for selected men; a drain is not routine.

Scenario	Catheter	Cystogram
Routine	~1 week, not before day 5	No
Extensive neck reconstruction / doubtful	≥10 days	Yes
Salvage	2–3 weeks	Strongly consider

- Removal at day 3–4 causes more retention. Extravasation delays but does not reduce long-term continence.
- **Penile rehabilitation**: PDE5 inhibitors help on-demand erections, but early rehabilitative use does not improve later response (3 RCTs). Use injections for non-responders.

9 · MIS-specific complications

Complication	Rate	Management
Transfusion	1–2%	Low, because pneumoperitoneum compresses veins
Bladder neck stricture	<2% (open up to 16%)	Dilatation or incision; rarely urethroplasty
Thromboembolism	~0.5%	Pneumatic compression; drugs only with multiple risk factors
Robotic failure	<0.5%	Cancel or convert
Recognised rectal injury	Rare; higher in salvage	Multilayer repair and flap, no colostomy
Unrecognised rectal injury	Rare	Presents as rectovesical fistula; colostomy, later repair
Urine leak	Uncommon	Drain, longer catheter, cystogram

- **Ureteric injury** occurs behind the vas (posterior approach), at the posterior bladder neck after TURP or with a median lobe, and during proximal PLND.

- With DVT, image the pelvis for lymphocele or haematoma; with a significant lymphocele, do a duplex for occult DVT.

> **KEY POINT**
> Robotic injuries are fewer but less visible: bleeding and strictures fall with magnified vision, while bowel injury and positioning neuropathy happen outside the camera's field and must be anticipated.

10 · Outcomes

- **Oncologic**: positive surgical margin (PSM) rates are equal open and robotic (~15%; ~9% organ-confined). PSM depends on tumour (Gleason ≥7, perineural invasion, stage) and surgeon volume, **not** on obesity, gland size or prior surgery.
- **Continence**: no-pad rates are in the mid-90s% at 1 year. Persistent mild leakage needs a sling or ProACT; severe (<2%) an artificial urinary sphincter.
- **Potency**: up to 90% at 1 year in expert hands; recovery continues to 5 years.
- **Salvage RALRP**: exclude metastases with PSMA PET and use full bowel prep and an air-bubble test. Continence is only 33–80%, and most men are impotent.

11 · Landmark trials

Trial	Design	Result	Take-home
Yaxley 2016	RCT, one open vs one robotic surgeon	Equal continence, potency, early oncology at 3 months; blood loss ~1400 vs 300 mL; shorter stay	Robot wins on recovery, not cancer or function
Retzius-sparing RCTs (Cochrane)	Single-centre RCTs	Earlier continence; equal at 12 months; more PSM	Use selectively

162 Radiation Therapy for Prostate Cancer

P2 P4 Radiotherapy and brachytherapy for CA prostate; Role of radiotherapy in focal and advanced CA prostate — α/β, schedules, brachytherapy, ADT by risk, salvage, Phoenix, toxicity

Not in the paper answer:

- A single 19 Gy high-dose-rate fraction fails (§5).
- RADAR and RTOG 99-10 on ADT duration (§6).
- Salvage prostatectomy incontinence ~63% (§11).

1 · Radiobiology and fractionation

- Conventional fractionation gives **1.8–2 gray (Gy)** daily.
- In the **linear–quadratic model**, the **α/β ratio** sets a tissue's response to fraction size: low (2–4 Gy) in late-responding normal tissue, high (≥8 Gy) in most tumours.
- Prostate cancer has few cycling cells, so its α/β is **≈1.5 Gy**, **lower than the rectum (~3 Gy)**. Fractions >2 Gy therefore favour tumour kill over late toxicity, which is the rationale for hypofractionation.

Category	Dose/fraction	Typical course
Conventional	1.8–2 Gy	~8 wk (78 Gy in 39 fractions)
Moderate hypofractionation	2.25–4 Gy	~4 wk (**60 Gy/20**)
Ultrahypofractionation = stereotactic body RT (SBRT)	≥5 Gy	≤5–7 fractions (**36.25 Gy/5**)

2 · EBRT techniques

- MRI is fused with planning CT; PSMA PET changes management in up to half of higher-risk men (see Ch 152).
- **IMRT or VMAT** lowers gastrointestinal (GI) but not genitourinary (GU) toxicity versus 3D conformal RT, as the urethra lies inside the target.

Volume	Prostate application
Gross tumour volume (GTV)	Dominant intraprostatic lesion (DIL) on MRI
Clinical target volume (CTV)	Prostate, with or without proximal seminal vesicles and pelvic nodes
Planning target volume (PTV)	CTV plus a margin for motion and set-up: **3–7 mm** with image guidance

- **Image-guided RT (IGRT)** with daily gold fiducials or cone-beam CT is the NCCN standard; its smaller margins make large fractions safe.
- A **rectal spacer** (hydrogel or hyaluronic acid) is injected transperineally into Denonvilliers' fascia. Late rectal toxicity falls to **2% vs 7%** (hydrogel, 79.2 Gy).
- The MR-linac allows 2 mm margins, and protons have shown no randomised benefit over IMRT.

3 · Dose escalation

- Six randomised trials, including RTOG 0126, raised **biochemical control in all risk groups** with **no overall survival (OS) gain**, because salvage works and metastasis fell only slightly. Still, ≥~78 Gy is standard.

- **FLAME**: a focal boost to the MRI DIL improved biochemical progression-free survival (HR 0.45) without more toxicity.

4 · Hypofractionation

- **Moderate**: CHHiP 57 Gy/19 was not shown noninferior; PROFIT 60/20 was noninferior; RTOG 0415 70 Gy/28 was noninferior but more toxic late; HYPRO 64.6 Gy/19 was **not superior**.
- After prostatectomy, HYPORT 62.5 Gy/25 was noninferior to 66.6 Gy/37.
- **SBRT**: HYPO-RT-PC gave ~84% 5-yr failure-free survival, with more acute toxicity.
- **Dose ceiling**: there is no gain above **40 Gy/5**. Give it on alternate days with 3–5 mm margins.

5 · Brachytherapy (BT)

Urinary function decides eligibility.

	Low-dose-rate (LDR)	High-dose-rate (HDR)
Source	Permanent seeds (^{125}I, ^{103}Pd, ^{131}Cs)	Temporary ^{192}Ir (380 keV) in a remote afterloader
Technique	TRUS-guided transperineal template; **peripheral loading** spares urethra	Dose shaped by dwell times and positions; no post-implant dosimetry
Uses	Monotherapy for low/intermediate risk; boost	Boost (e.g. 15 Gy × 1 after 46 Gy); monotherapy 27 Gy/2, 34–38 Gy/4
Pitfall	Oedema, seed migration	**19 Gy × 1 fails**: 5-yr local failure 29% vs 3% (27 Gy/2)

Isotope	Half-life	Monotherapy	Boost (with 45 Gy EBRT)
^{125}I	**59.6 d**	140–160 Gy	108–110 Gy
^{103}Pd	17 d	110–125 Gy	90–100 Gy
^{131}Cs	9.7 d	100–115 Gy	80–85 Gy

- **Post-implant dosimetry** (CT at 14–30 days): **D90** >90% of prescription; **V100** >85%; urethral V150 <0.05 cm³; rectal V100 <1 cm³. A low D90 predicts biochemical failure. MCQ D90 is a dose, V100 a volume.
- **Relative contraindications**: **IPSS >20**; **Qmax <10 mL/s**; large median lobe; large TURP defect or TURP within 3–6 mo; prior pelvic RT; inflammatory bowel or connective tissue disease; **pubic arch interference**; inability to adopt lithotomy.
- A large gland shrinks ~30% with ADT. Retention risk is ≈29% if IPSS >20 vs ≈2% if <10, so start α-blockers before the implant.
- **With EBRT**: 45 Gy plus an implant at 60–70% of monotherapy dose. **RTOG 0232**: BT alone equalled EBRT plus BT in intermediate risk with less toxicity, so BT alone is standard.
- **ASCENDE-RT**: no OS difference; LDR boost grade 3 GU toxicity **18.4% vs 5.2%**.

KEY POINT

Why BT works: the steep fall-off lets intraprostatic dose far exceed EBRT. Hence outcome depends on implant accuracy (D90), eligibility on whether the urethra tolerates swelling, and EBRT is added only where disease may lie beyond the implant.

6 · ADT with RT

- **Synergy**: ADT causes apoptosis, shrinks the target and lowers **Ku70**, impairing non-homologous end-joining repair (see Ch 164).

Risk group (NCCN)	With EBRT	With EBRT + BT	Key evidence
Low / favourable intermediate	None	—	RTOG 94-08: no benefit in low risk
Unfavourable intermediate	**4–6 mo**	—	RTOG 94-08, DFCI 95-096, TROG 96.01 (6 mo better than 3); RTOG 99-10: 9 mo no better than 4
High / very high	**24–36 mo** (≥18)	12–36 mo	EORTC 22863, RTOG 92-02, EORTC 22961 (36 mo better than 6); PCS IV: 18 mo = 36 mo

- Needed duration falls as RT dose rises: in **RADAR**, 18 mo beat 6 mo only without an HDR boost.
- **STAMPEDE**: adding 2 yr abiraterone in high-risk N0/N1 M0 improved metastasis-free survival and OS.

7 · Pelvic nodal RT

- RTOG 94-13 and GETUG-01 showed no durable benefit, mainly because a field border at L5/S1 misses the **common iliac nodes**; modern fields extend to **L4/L5**.
- **POP-RT** (node risk >20%, long ADT): whole-pelvis RT raised 5-yr biochemical failure-free survival (~95% vs ~81%), with no OS difference and more acute GI toxicity.

8 · Post-prostatectomy RT

- RADICALS-RT, GETUG-AFU 17 and RAVES, with their meta-analysis, make **early salvage preferred** over adjuvant RT, which may still suit very high risk (Gleason 8–10, pT3b) (see Ch 164).
- Dose is ≥64 Gy, and PSMA PET localises recurrence.

9 · Follow-up and biochemical failure

Do not mistake a benign bounce for failure.

- Benign glands persist, so PSA stays detectable. The nadir is ≤0.1 ng/mL after EBRT plus BT and higher after EBRT alone; it is reached at ~1.5 yr after EBRT and ~2.5 yr after BT.

	ASTRO 1996	Phoenix 2005 (current)
Failure	3 consecutive rises	PSA **≥ nadir + 2 ng/mL**
Date	Backdated to midway between nadir and first rise	When reached (or when salvage started)
Notes	Delays diagnosis ≥18 mo; invalid with ADT or BT	Best correlates with clinical relapse; valid with short ADT; fewer false calls from bounces

- Poor prognosis is signalled by failure <18 mo and especially **PSA doubling time <3 mo**, which predicts cancer death.
- A **bounce** (rise >0.2 ng/mL, then durable fall) occurs in ~1/3 after LDR, mostly in the first 3 yr, and predicts a better outcome.
- **Biopsy** is best at **30–36 mo**, since irradiated cells die slowly. Marked radiation effect is not graded and behaves as negative.

10 · Toxicity

Toxicity	Features	Management / prevention
Acute GU	Peaks ~2 mo, settles by 12 mo	α-blockers; catheter for retention
Late GU	Grade ≥2 ~20%; stricture; incontinence ~3%, more after TURP	IGRT halves late GU toxicity; peripheral loading
Late GI (proctitis)	Bleeding at 6–18 mo; IMRT 81 Gy grade ≥2 ~5%; BT 1–2%; colostomy ~0.3%	Steroid or sucralfate enemas, then **argon plasma coagulation** or hyperbaric oxygen, then colostomy
Erectile dysfunction	Vascular, over 24 mo; often due to ADT; best with BT alone (~76% potent at 5 yr)	PDE5 inhibitors work in ~¾
Second cancers	Small excess of bladder and rectal cancers	Investigate haematuria or rectal bleeding fully

11 · Salvage after local failure

- Salvage needs **biopsy-proven, local-only** recurrence. Options are salvage RP (incontinence ~63%), BT (preferred), SBRT re-irradiation, cryotherapy and HIFU.

12 · Landmark trials

Trial	Design	Result	Take-home
RTOG 0126	70.2 vs 79.2 Gy, low–intermediate	8-yr biochemical failure-free 65% vs 80%; OS same	Dose escalation improves control, not OS
CHHiP	74/37 vs 60/20 vs 57/19	60/20 noninferior	60 Gy/20 standard
HYPO-RT-PC	42.7/7 vs 78/39	Noninferior; equal late toxicity	Ultrahypofractionation valid
PACE-B	36.25/5 vs conventional/moderate	Similar 2-yr toxicity	SBRT safe
ASCENDE-RT	EBRT vs LDR boost (plus 12 mo ADT)	Biochemical failure HR 2.04 for EBRT boost; more GU toxicity with LDR	BT boost for high risk
TROG 96.01	RT with no, 3 or 6 mo ADT	6 mo improved OS; 3 mo did not	≥6 mo short-course
RTOG 94-08	66.6 Gy with or without 4 mo ADT, T1b–T2b	10-yr OS 57% vs 62%; intermediate risk only	ADT for intermediate, not low
EORTC 22863	RT with or without 3 yr goserelin, T3–T4	10-yr OS 39.8% vs 58.1%	Long ADT in high risk
EORTC 22961	6 vs 36 mo ADT	Short course not noninferior	Long ADT for high risk

Trial	Design	Result	Take-home
PCS IV	18 vs 36 mo	No survival difference	18 mo acceptable
SPCG-7; NCIC PR.3/MRC PR07	Lifelong ADT with or without RT	RT lowered cancer death and raised OS	Never ADT alone if fit
RTOG 94-13; GETUG-01	Pelvic vs prostate-only RT	No durable benefit	Field height matters
SPPORT	Salvage bed RT, with or without ADT and nodes	Pelvic RT plus ADT best	Treat nodes in salvage

163 Focal Therapy of Prostate Cancer

1 · Definition and rationale

- **Focal therapy (FT)** is guided ablation of an image-defined, biopsy-confirmed lesion with a **safety margin**. Sparing the neurovascular bundles (NVBs), sphincter and rectum preserves function.
- FT rests on mpMRI, targeted biopsy and the fact that **Gleason pattern 4/5** is lethal.

Index lesion hypothesis

- Cancer is multifocal in **50–91%** of radical prostatectomy (RP) specimens. The **index lesion (IL)**, the largest or highest-grade focus, is held to drive natural history and metastasis.
- **For**: the IL holds **80–90%** of tumour volume, and satellites are mostly **< 0.5 mL** (Stamey "insignificant").
- **Against**: a satellite is the aggressive focus in 12–38%, and IL Gleason matches overall grade in only ~⅔.

> **KEY POINT**
> "Multifocal, therefore unsuitable for FT" is a trap: what matters is one significant lesion plus insignificant satellites, not the count of foci. The hypothesis's exceptions are why FT is always paired with surveillance.

- **FT and active surveillance (AS)** form one strategy: ablate grade 4 disease, watch the low-grade remainder, and retreat new foci.

2 · Patient selection

- The patient must **accept close surveillance**.
- **TRUS biopsy** misses almost half the disease relevant to hemiablation compared with **transperineal mapping biopsy (TMB, 5-mm grid)**.
- **mpMRI** detects high-grade and anterior tumours. In **PROMIS**, its sensitivity for clinically significant cancer (csPCa) was **93%** and specificity 41%, so MRI lesions need biopsy and MRI-negative areas still need systematic biopsy.
- mpMRI **underestimates volume**, so a margin is needed: **5 mm** ablates >95% of csPCa. About **90%** of outfield csPCa lies within **1 cm** of the target.

> **KEY POINT**
> Imaging finds the lesion, only biopsy grades it, and only systematic biopsy vouches for the rest of the gland. Every consensus has kept all three.

Consensus criteria (2016–2019)

Variable	Consensus
Work-up	mpMRI plus targeted and systematic biopsy
Risk group	**Intermediate**; Gleason **3+4 = "sweet spot"**; 4+3 maximum; no cribriform or ductal histology
Size	≤1.5 mL (≤3 mL if not crossing midline); 20–25% of gland if crossing
Residual untreated disease	Gleason 3+3 only (≤1 mm core length in 2016, tightened from 5 mm)

Variable	Consensus
Life expectancy	Per guidelines; no fixed limit
Erectile dysfunction	**Not** a contraindication
Not suitable	High risk (no long-term data); low risk usually goes to AS

- The goal is now to treat and retreat so as to **delay or avoid radical therapy**. FT remains investigational, ideally within a trial.

3 · Ablation patterns and energies

- FT means **lesion, quadrant or hemiablation**; "hockey-stick" (subtotal) ablation is partial-gland ablation, not FT.

P2 P4 HIFU; Cryotherapy for carcinoma prostate — HIFU ≥55 °C, posterior lesions; cryo Joule–Thomson, urethral warmer, fistula ≤0.8%

Not in the paper answer:

- Focal cryotherapy 5-yr biochemical progression-free survival (BPFS) ~62–70%; COLD registry: partial equals whole-gland control with ~2× better erections.

Modality	Mechanism / route	Pros	Cons
TULSA (MRI-guided transurethral ultrasound ablation)	Transurethral transducer with MR thermometry	Anterior gland; control within 1.3 mm	In-bore MRI; csPCa on 12-month biopsy ~⅓
Focal laser	Interstitial fibre ≥60 °C; MR thermometry	Most predictable zone	Narrow margin, so **highest infield failure** (~40% at 6 yr)
Irreversible electroporation (IRE)	Non-thermal pulses >1600 V/cm open membrane pores	Spares connective tissue; good function	Needs deep anaesthesia; NVB and urethral injury on histology
Vascular-targeted photodynamic therapy (VTP)	IV padeliporfin (TOOKAD) 4 mg/kg activated by 753-nm light occludes vessels	**Only randomised evidence** (PCM301)	Gleason 6 only; perineal pain ~15%; light precautions
Radiofrequency	~100 °C	Fat limits conduction	Infield csPCa 20–30%
Focal brachytherapy	Low-dose-rate seeds or high-dose-rate (HDR) catheters	Mild early toxicity	HDR: stricture ≤8.2%, ED in half; biopsy only at 24 mo

KEY POINT

Every energy fails at the edge of the ablation zone, where temperature or field strength becomes sublethal. Margins, thermometry and adjuvants all aim to make that edge lethal, so laser's precise but narrow zone gives the most infield recurrence.

4 · Outcomes

Modality	Infield positive biopsy	Outfield	Continence	Erections
Cryotherapy	7–39% overall (~¼ csPCa)		95–100%	47–100%
HIFU	17–54% (csPCa 0–22%)	7–43% (csPCa 2–8%)	90–95%	50–78%

Modality	Infield positive biopsy	Outfield	Continence	Erections
Laser	11–50%	17–60%	95–100%	Minimal change
IRE	csPCa 1–10%	csPCa 5–16%	~All	Some decline
VTP	19–37%	17–19%	Preserved	Nil to mild fall

- HEAT (HIFU): 7-yr failure-free survival was **69%** (68% intermediate, 65% high risk).

5 · Complications

- The commonest is **urinary retention** (0–9%); others are infection, sloughing and stricture (~4–5% after HIFU or IRE).
- **Rectourethral fistula** occurs in ≤2% and is the most feared; some heal with a catheter for 4–6 weeks.

6 · Follow-up and failure

Follow-up relies on MRI and biopsy, not PSA.

- **PSA** falls only **50–80%**, and the Phoenix/ASTRO definitions are not validated after FT. A rise to nadir + **1.0** ng/mL at 12 months, or + 1.5 at 24–36 months, predicts csPCa.

Success (ICUD–SIU 2015)	Infield	Outfield
Short term	Eradication of treated focus	No csPCa — any outfield cancer = **selection failure**
Intermediate/long term	No csPCa	No csPCa (new cancer may be de novo)

ALGORITHM

1. At **3–6 months**: mpMRI and targeted biopsy (4–6 cores) of the treated zone.
2. At 1–2 and 5 years: mpMRI; if a new lesion appears, then targeted biopsy.
3. At 1–2 and 5 years: systematic biopsy of the untreated gland.

- **PI-FAB**: read contrast-enhanced (**DCE) images first**, as scar obscures T2 and diffusion. Score 1 (both negative) is monitored, score 2 (DCE positive only) is biopsied if PSA rises, and score 3 (both positive) is biopsied.

7 · Salvage

- After FT failure, options are **focal retreatment** or whole-gland RP or radiotherapy.

Focal salvage for radiorecurrent disease

	Cryotherapy	HIFU	IRE
Cancer control	5-yr BPFS ~50%	3-yr BPFS 48%	Progression-free survival 60%, metastasis-free survival 91% (48 months)
Continence	87–100%	64–87.5%	93%
Signature complication	Rectourethral fistula ~1.4%	Bladder-neck stricture, fistula ~2%; grade ≥3 ~¼	Urethral sludge ~9%

- Focal salvage controls cancer as well as whole-gland salvage, with better function.

8 · Landmark trials

Trial	Design	Result	Take-home
PCM301 (Azzouzi 2017)	415 low-risk men; padeliporfin VTP vs AS	Progression 28% vs 58%; conversion to radical therapy at 4 yr **24% vs 53%**; continence preserved	First level 1 evidence for FT — but Gleason 6 only, surrogate end points
PROMIS	mpMRI vs 5-mm TMB, 11 centres	csPCa sensitivity 93%, specificity 41%	MRI lesions need biopsy; systematic biopsy still needed
HEAT registry	Prospective focal HIFU, 1379 men	7-yr failure-free survival 69%	Largest FT dataset; includes high risk

KEY POINT

Open problems are leaving satellites, the cost of retreatment and adjuvants. Lumpectomy works because the breast gets adjuvant radiotherapy, whereas prostate FT relies on surveillance and retreatment.

164 Treatment of Locally Advanced Prostate Cancer

P2 P1 Staging and management of locally advanced prostate cancer; Staging and management of non-organ-confined CA prostate — definition, risk groups, work-up, RT + ADT, RP, salvage RT, pN1, algorithm

Not in the paper answer:

- pN1: hormones plus RT beat hormones alone (OS 74% vs 55%) (§3).
- Bicalutamide 150 mg raised death in localised disease (HR 1.23) (§5).

1 · Definition and natural history

- The working definition is **T3–4, N0 or N1, M0**, and it calls for **multimodal therapy**.
- NCCN high risk is cT3a, PSA >20 or grade group (GG) 4–5. Very high risk is cT3b–T4, primary pattern 5, ≥2 high-risk features, or >4 cores of GG 4–5.
- Because of stage migration, high risk now mostly means **high grade**, not a palpable T3.
- **Grade drives death**: untreated Gleason 8–10 had 15-yr cancer mortality of 60–87% before the PSA era.

2 · Staging and risk assessment

- Clinical staging and imaging both **understage**.
- NCCN orders a bone scan and CT or MRI for cT3–T4, PSA >20, GG 4–5 or primary pattern 5; PSMA PET may substitute.
- In **proPSMA**, PSMA PET accuracy was **92% vs 65%**.

3 · Radical prostatectomy in high-risk disease

- **Clinical cT3 is overstaged in 7–27%**, and >⅓ of high-risk specimens are confined, so do not exclude men on clinical stage alone.
- NCCN advises RP with extended pelvic node dissection (**ePLND**) only if the tumour is **completely excisable**; otherwise give androgen deprivation (ADT, "AD" in trial reports) with radiotherapy (RT).
- In cT3, DSS after RP is **~80–90% at 10 yr**.

> **KEY POINT**
> Operative principle: a **positive surgical margin** carries the highest relative risk of progression (4.37), higher than extracapsular extension (ECE), seminal vesicle invasion (SVI) or nodes, so remove all prostatic tissue regardless of stage.

Pathology after RP	5-yr PSA-free survival	Relative risk of progression
Organ-confined	95%	Reference
ECE	76%	2.2–2.7
SVI	37%	2.6
Node-positive	18%	3.3
Positive margin	—	**4.37** (highest)

Neoadjuvant AD before RP

- **Clinical downstaging occurs in 32–90%, but pathological in only 8–31%**. It has **no** effect on PSA recurrence, DSS or overall survival (OS).

- **CALGB 90203** (AD plus docetaxel before RP) missed its primary endpoint but improved MFS and OS.

Adjuvant and salvage RT after RP

Trial	Population / comparison	Result
EORTC 22911	pT3 and/or positive margin; immediate RT vs observation	Biochemical disease-free survival 74% vs 53%; **benefit only with positive margins**
SWOG 8794	Same	**MFS (HR 0.71) and OS (HR 0.72)** — the only OS-level evidence
ARO 96-02	pT3N0, undetectable PSA	Biochemical HR 0.53
RADICALS-RT, GETUG-AFU 17, RAVES	Adjuvant vs early salvage (64–66 Gy)	**Early salvage preferred**; 5-yr event-free 87–94% in both arms

- **Trap**: very high-risk men (Gleason 8–10, pT3b) were only ~20% of the timing trials, so adjuvant RT may still help them, as may extensive margins.
- Dose is **≥64 Gy**. **GETUG-AFU 16**: **3 months** of AD with salvage RT gave 10-yr progression-free survival 64% vs 49% (HR 0.54) (see Ch 162).

Adjuvant AD and pN1

- In **pT3N0**, adjuvant AD reduces progression but gives **no DSS or OS gain**.
- **ECOG 7887** (nodes found at RP): immediate AD improved OS, DSS and PFS.
- **AD plus RT beats AD alone** in pN1: 10-yr DSS 86% vs 70%, OS 74% vs 55%.

> **ALGORITHM**
>
> 1. **pN1 after RP**
> 2. If favourable (few nodes, Gleason ≤8), then observe with PSA monitoring.
> 3. If adverse (>3 nodes, Gleason >8), then give AD plus RT (best OS).
> 4. Avoid AD alone in a fit man.

4 · RT + ADT

- The standard is **dose-escalated RT (>72, ideally ~78–81 Gy) with ADT**, often with pelvic RT (see Ch 162).
- ADT shrinks the target and is synergistic with RT.

Trial	Question	Verdict
RTOG 86-10	4 mo neoadjuvant/concurrent AD, bulky T2–T4	Disease-specific mortality fell from 36% to 23%; **OS unchanged**
TROG 96.01	RT with no, 3 or 6 mo NAD	6 mo lowered all-cause mortality (HR 0.63); 3 mo did not
RTOG 85-31	Indefinite adjuvant goserelin	Better control and MFS; OS gain only in Gleason 8–10
PCS IV	18 vs 36 mo	OS comparable; 18 mo better quality of life
RTOG 94-13	Pelvic vs prostate RT, and neoadjuvant vs adjuvant AD	Early PFS gain lost; whole pelvis with adjuvant AD had worst OS

- **Duration**: intermediate risk 4–6 mo; high risk (NCCN) **18–36 mo** with external beam RT (EBRT), 12–36 mo with a brachytherapy boost.

- **ASCENDE-RT**: a low-dose-rate boost halved biochemical recurrence vs an EBRT boost (HR 2.04), with equal OS.
- **Trap**: AD plus RT benefits often exclude OS. The OS gain is seen with long AD in high risk (EORTC 22863; RTOG 92-02 Gleason 8–10 OS 81% vs 71%).

Adding RT to ADT

- **SPCG-7**: cancer death relative risk **0.44**; overall mortality RR 0.68. **PR.3/PR07**: DSS HR 0.54, and 7-yr OS rose from 66% to 74%. So RT adds **8–10% OS**.

Systemic intensification

Trial	Added agent	Verdict
RTOG 0521	Docetaxel to long-term AD and RT	OS HR 0.69
GETUG 12	Docetaxel and estramustine to AD	Relapse-free 62% vs 50%
RTOG 9902	Paclitaxel, estramustine and etoposide	No benefit; thromboembolism
STAMPEDE (high-risk M0/N1)	**Abiraterone and prednisolone for 2 yr**, added to 3 yr AD and RT	**MFS HR 0.53, OS HR 0.60**

- STAMPEDE eligibility: **N1, or two of Gleason ≥8, cT3–4 and PSA ≥40**. **NCCN 2023 advises RT, ADT and 2 yr abiraterone for very high risk.**

5 · ADT alone and other options

- **MRC**: immediate rather than deferred AD gave fewer cancer deaths, mainly in M0.
- **Bicalutamide 150 mg** improved PFS in locally advanced disease but **raised death in localised disease** (HR 1.23), so never use it at low risk.
- AD alone does not stop local progression, and over half need a TURP.
- **RP vs RT with ADT**: there is no completed randomised trial; **SPCG-15** (cT3N0M0) is ongoing.

ALGORITHM

1. **Locally advanced / high-risk M0**
2. Stage with PSMA PET, or CT and bone scan.
3. If fit and resectable, then RP with ePLND. If pathology is adverse, then early salvage RT (adjuvant if pT3b, Gleason 8–10 or positive margins). If pN1, then observe or give AD plus RT.
4. Alternative: dose-escalated RT (with or without brachytherapy boost and pelvic nodes) plus 18–36 mo ADT.
5. If very high risk or N1, then add 2 yr abiraterone.
6. If unfit for local therapy, then ADT alone.

KEY POINT

Viva: "Why not ADT alone for T3 disease?" SPCG-7 and PR.3 show RT added to ADT halves cancer death and improves OS; AD alone neither cures nor controls the primary, and it carries its own toxicity.

6 · Landmark trials

Trial	Design	Result	Take-home
SPCG-7	ADT with or without RT, mostly T3	Cancer death RR 0.44	Add RT to ADT
NCIC PR.3/MRC PR07	Lifelong ADT with or without RT	OS HR 0.77	Add RT to ADT
EORTC 22863	RT with or without 3 yr goserelin	5-yr OS 62% vs 79%	Long ADT in high risk
RTOG 92-02	4 vs 28 mo ADT	OS gain in Gleason 8–10	Prolong ADT for high grade
STAMPEDE (abiraterone)	ADT and RT, with or without 2 yr abiraterone	MFS HR 0.53, OS HR 0.60	Standard for very high risk/N1
SWOG 8794	Adjuvant RT vs observation	MFS and OS improved	Postoperative RT works
RADICALS-RT / RAVES / GETUG-AFU 17	Adjuvant vs early salvage	No difference	Early salvage for most
ECOG 7887	pN+ immediate vs delayed AD	OS improved	Treat pN+ early
proPSMA	PSMA PET vs conventional imaging	Accuracy 92% vs 65%	PSMA PET for staging

165 Management of Recurrent and Newly Metastatic Prostate Cancer

1 · Biochemical recurrence: concept and definitions

Biochemical recurrence (BCR) is a rise in prostate-specific antigen (PSA) after curative therapy, before clinical recurrence. Clinical recurrence is almost always preceded by it.

- After radical prostatectomy (RP), **¼–⅖** develop BCR within 10 years; only ≈half of higher-risk men with BCR fail clinically.
- At MSK, PSA **≥0.4 ng/mL followed by a further rise** best predicted metastasis.
- **Ultrasensitive PSA** is prognostic, not diagnostic; about half of early values fall on retesting, so repeat it.

Setting	Definition	Logic
Post-RP — AUA	PSA **≥0.2 ng/mL**	No gland remains, so any PSA is suspect
Post-RP — NCCN	Never undetectable, two rises, or PSA ≥0.1	Any of three triggers
Post-RP — EAU	No threshold; risk groups (below)	Kinetics plus grade
After external beam RT (EBRT) ± androgen deprivation therapy (ADT)	**Phoenix**: **nadir + 2 ng/mL**	Validated against metastasis and death
After low-dose-rate (LDR) brachytherapy	≥0.2 may suit better; **<0.2 at 4 years ≈ cure**	The very low nadir means Phoenix may flatter brachytherapy

EAU BCR risk group	After RP	After radiotherapy (RT)
Low risk (needs BOTH)	PSA doubling time (PSADT) **>1 year** AND grade group (GG) <4	Interval to failure **>18 months** AND GG <4
High risk (EITHER)	PSADT ≤1 year OR GG 4–5	Interval ≤18 months OR GG 4–5

KEY POINT

PSA bounce after RT is a transient rise of 0.1–0.5 ng/mL at a mean of ≈ **9 months**, followed by a fall. It is **favourable**, so monitor; never salvage on a bounce.

2 · Risk prediction and natural history

- A **positive margin** is the only modifiable BCR predictor.
- **CAPRA** is the pre-operative score and **CAPRA-S** the post-operative one MCQ. Only the MSK nomograms predict 15-year cancer-specific survival.
- **Pound**: untreated, a median **8 years** passes from BCR to metastasis, then **5 years** to death (P2-144).
- After Phoenix failure, metastasis follows at ≈5.4 years. Follow-up is lifelong, because ¼ recur after 5 years.

3 · Imaging at recurrence

P2 65 yr male underwent RP… PSA has risen — BCR definitions, PSMA PET/mpMRI work-up, salvage RT dose/timing, RTOG 9601, GETUG-AFU 16, SPPORT, STOMP/ORIOLE, EMBARK, Pound prognosis

- Bone scan and CT are rarely positive below PSA **10 ng/mL**.
- **Prostate-specific membrane antigen (PSMA) PET** is positive in ≈½ at PSA **<0.5**, and finds disease outside the salvage field in ≈1 in 5. Fluciclovine-guided salvage improved control (**EMPIRE-1**).
- Multiparametric MRI (mpMRI) is best for local recurrence. **Biopsy after RT** is mandatory before local salvage.

> **KEY POINT**
> Salvage RT works best at the lowest PSA, but imaging improves only as PSA rises, so risk features (grade, margins, kinetics) guide early action.

4 · Salvage after radical prostatectomy

- **Salvage RT (SRT)** is the only curative option (P2-144). Give **≥64–65 Gy** to the fossa.
- Delay costs control: ≈10% relapse-free survival is lost per 0.1 ng/mL in men with ≥2 adverse features.
- No visible lesion on mpMRI predicts SRT failure, because the disease is probably occult metastasis.

5 · Salvage after radiotherapy

Prerequisites are biopsy-proven local recurrence, no metastasis on imaging, excellent performance status, life expectancy ≥5–10 years and a shared decision. Every modality gives ≈**50–70%** control, so selection (local-only disease, low PSA, GG <4) matters more than modality.

Option	Selection / features	Signature morbidity
Salvage RP (open/robotic)	EAU: life expectancy ≥10 years, PSA **<10**, biopsy GG ≤3, N0M0 on imaging, initial cT1–T2.	Incontinence, bladder neck contracture, rectal injury, near-universal impotence
Salvage brachytherapy (LDR ≈ high-dose-rate)	Only after EBRT	Grade ≥3 urinary toxicity ≈1 in 6
Salvage stereotactic body RT (SBRT)	After EBRT	Low toxicity, modest control
Salvage cryotherapy	PSA <10 (ideally <4), PSADT **≥16 months**; a post-cryo nadir <0.2 predicts success	**Urethrorectal fistula** (1–3%), erectile dysfunction; incontinence single-digit with **urethral warming**
Salvage high-intensity focused ultrasound (HIFU)	≥2 years after EBRT	Fistula, osteitis pubis, stricture, incontinence

Recurrence after **focal therapy** is worked up like RT failure.

6 · Non-curative approach to BCR: observation vs ADT

- AUA: observe (or enrol in a trial) after maximal local therapy; give ADT only for higher metastatic risk (PSADT **<12 months** or high grade), preferably intermittent.
- EAU advises observing low-risk BCR, limited life expectancy, or men declining local therapy. Primary ADT for localised disease is **not** recommended.
- **Messing (ECOG)**: immediate ADT for node-positive (pN+) disease after RP improved overall survival (OS), but the trial was small and pre-dated modern imaging.
- Low-volume pN+ with undetectable PSA: **20–40%** are cured by surgery alone.
- Immediate ADT: TOAD OS hazard ratio (HR) 0.55; EORTC 30891 benefit only if PSA >50 or PSADT <12 months.
- **EMBARK** entry: PSADT **≤9 months**, with PSA ≥1 after RP or ≥2 above nadir after RT.

Intermittent vs continuous ADT

- **NCIC PR.7** (rising PSA after RT): intermittent was **non-inferior** (OS 8.8 vs 9.1 years).
- **SWOG 9346** (new metastatic): OS 5.1 vs 5.8 years, HR 1.10 (90% CI 0.99–1.23). This was **inconclusive**; intermittent may be worse here.

7 · Androgen deprivation therapy: biology and agents

Castration removes the ligand, not the receptor, so castration-resistant prostate cancer (CRPC) keeps androgen receptor (AR) signalling alive through AR amplification, ligand-binding-domain mutations, splice variants (**AR-V7**, poor prognosis) and tumour steroid synthesis. **Castrate testosterone: <50 ng/dL**.

P4 P2 GnRH (LHRH) analogues; Maximum androgen blockade and bicalutamide — agonist flare/FSH rise, antagonists (degarelix, relugolix/HERO), antiandrogen toxicities, CAB, withdrawal response

Method	Key facts
Bilateral orchiectomy	Castrate within hours. Fewer cardiac events and fractures than agonists with equal survival, so offer it for lifelong ADT
LHRH agonists / antagonists	See P4-103. HERO: castrate by day 4 in 56% vs 0%
Cyproterone acetate	Inferior to goserelin; cardiovascular events up to 10%; used 50–100 mg/day for hot flushes
AR-targeting agents (ARTAs): enzalutamide 160 mg, apalutamide 240 mg, darolutamide 600 mg twice daily	Seizures with enzalutamide; rash and hypothyroidism with apalutamide; darolutamide has the fewest effects
Abiraterone (selective CYP17), ketoconazole (non-selective)	Give abiraterone with prednisone to prevent ACTH-driven mineralocorticoid excess. Ketoconazole castrates within 4 h: an emergency option

KEY POINT

Flare matters with spinal or weight-bearing bone metastases: give an antiandrogen with the agonist for **21–28 days**. For **impending cord compression**, use an **antagonist or orchiectomy** (or ketoconazole). **Antiandrogen withdrawal**: 15–30% respond with bicalutamide and 0–10% with ARTAs, with no OS gain.

8 · ADT side effects and their management

Effect	Key facts	Management
Osteoporosis / fracture	Loss of peripheral oestrogen; fracture risk rises 20–45%	Baseline DEXA and FRAX. For all: calcium 1000–1200 mg and vitamin D 400–1000 IU/day, exercise, stop smoking. Give a drug if T-score is −1 to −2.5 with 10-year hip fracture risk **>3%** or major fracture **>20%** (or osteoporosis): denosumab 60 mg SC 6-monthly, zoledronic acid 4–5 mg IV yearly or alendronate 70 mg weekly. Repeat DEXA at 1 year
Hot flushes (commonest, 50–80%)	Hypothalamic oestrogen loss	Oestrogens best; megestrol, cyproterone, venlafaxine, gabapentin. Clonidine is ineffective
Metabolic	Fat rises with sarcopenia; diabetes (agonist HR 1.44)	Exercise; monitor glucose and lipids
Cardiovascular	Agonists raise coronary events, sudden death and thromboembolism	Risk-factor control; orchiectomy or antagonist favoured
Anaemia	Normocytic; pre-ADT Hb <12 g/dL predicts shorter survival	Reversible ≈1 year after stopping
Gynaecomastia	≈½ with bicalutamide 150 mg or enzalutamide; ≤5% with castration	Prophylactic breast RT 10–15 Gy or tamoxifen

Follow-up on ADT

Check PSA and bloods 3–6-monthly. **If PSA rises, check testosterone first.** A rising PSA with castrate testosterone means **CRPC** (see Ch 166), so restage. The germline HSD3B1 (1245A>C) variant predicts ADT resistance MCQ.

9 · Metastatic hormone-sensitive prostate cancer (mHSPC)

P3 Management of newly Metastatic prostate cancer — CHAARTED/LATITUDE definitions, doublets and triplets with HRs, prostate RT for low volume, MDT, genetic testing, follow-up

Not in the paper answer:

- **SWOG 9346 PSA nadir at 7 months**: ≤0.2 gives median OS **75** months; 0.2–4 gives 44; >4 gives 13.
- Metachronous low volume does best (median OS ≈92 vs ≈43 months for synchronous high volume).
- Genes to test: germline BRCA1/2, ATM, PALB2, CHEK2, HOXB13, mismatch repair; somatic CDK12, MSI-H.

Definition	Criteria	Use
CHAARTED high volume	**≥4 bone metastases with ≥1 outside the axial skeleton**, OR any visceral metastasis (no grade criterion)	Docetaxel benefit
LATITUDE high risk	**≥2 of**: Gleason ≥8, ≥3 bone lesions, visceral	Abiraterone entry

- **ADT is the floor, never the ceiling**; long-term combined blockade is not recommended.

- Docetaxel is **75 mg/m² 3-weekly × 6** (meta-analysis HR 0.77).
- Observe only if asymptomatic with life expectancy <5 years. Cytoreductive prostatectomy is for trials only.
- **Metastasis-directed therapy (MDT)**: oligometastases are **≤3 sites** on PET, treated by SBRT or salvage node dissection.

10 · Landmark trials

Trial	Design	Result	Take-home
RADICALS, RAVES, GETUG-AFU 17 (ARTISTIC)	Adjuvant vs early salvage RT (PSA 0.1–0.2)	Equivalent; ≥60% spared RT	Early salvage preferred
RTOG 0534 (SPPORT)	Bed vs bed plus 4–6 months ADT vs pelvis plus ADT	5-year freedom from progression 70.9 / 81.3 / **87.4%**	Add ADT and pelvic nodes (not OS)
RTOG 9601	SRT ± 2 years bicalutamide 150 mg	12-year OS 76.3 vs 71.3%	OS benefit
GETUG-AFU 16	SRT ± 6 months goserelin	10-year BCR-free 64 vs 49% (HR 0.54); MFS HR 0.73; OS unchanged	Short ADT: no OS gain
EMBARK	High-risk BCR; leuprolide ± enzalutamide vs enzalutamide	MFS 87.3% (combination) / 80.0% (enzalutamide) / 71.4% (leuprolide); HR 0.42 and 0.63	Enzalutamide for high-risk non-metastatic HSPC
CHAARTED (ECOG 3805)	ADT ± docetaxel	OS 57.6 vs 44.0 months (HR 0.61); high volume **49.2 vs 32.2** months	High volume only
GETUG-AFU 15	ADT ± docetaxel	OS HR 0.88, not significant	Negative docetaxel trial
STAMPEDE (docetaxel; arm G abiraterone)	Multi-arm; metastatic and high-risk M0	Docetaxel OS HR 0.78; abiraterone OS HR 0.63, failure-free survival HR 0.29	Both work
LATITUDE	High-risk mHSPC; ADT ± abiraterone	OS HR 0.62; rPFS 33.0 vs 14.8 months	Abiraterone in high risk
ARCHES	ADT ± enzalutamide	rPFS HR 0.39	Volume-independent
ENZAMET	ADT plus enzalutamide vs ADT plus older antiandrogen	OS HR 0.67	Control was not placebo
TITAN	ADT ± apalutamide	rPFS HR 0.48; OS HR 0.67	Volume-independent
PEACE-1	Synchronous; 2×2, ± abiraterone and prostate RT	All: rPFS HR 0.54, OS HR 0.82; with docetaxel: rPFS 0.50, OS 0.75	Triplet with abiraterone
ARASENS	ADT and docetaxel, ± darolutamide	OS HR 0.68	Darolutamide in mHSPC only with docetaxel
STAMPEDE arm H	ADT ± prostate RT	Overall HR 0.92; low volume HR 0.68 (3-year OS 81 vs 73%)	RT to prostate in low volume

166 Management of Castration-Resistant Prostate Cancer

1 · Definition and biology

CRPC means progression (biochemical, radiographic or clinical) despite castrate testosterone **<50 ng/dL**. On androgen deprivation therapy (ADT) alone, metastatic overall survival (OS) was historically 24–36 months.

P2 Castration-resistant prostate cancer (CRPC): definition and treatment — EAU definition, AR mechanisms, evaluation, general principles, nmCRPC, mCRPC options table, sequencing, supportive care

Not in the paper answer:

- A prostate-specific antigen (PSA) doubling time (PSADT) **<3 months** predicts a rapid course, so consider aggressive therapy.
- Bone pain or visceral-predominant disease calls for **upfront chemotherapy**.
- **Castration resistance is not androgen independence.** The androgen receptor (AR) still drives growth through amplification, mutations, the splice variant **AR-V7** and tumour steroid synthesis, so AR-pathway inhibitors (ARPIs) work.
- Once AR dependence is lost, use taxanes, radium-223, PSMA therapy, PARP inhibitors (PARPIs) or immunotherapy.

2 · Clinical states and progression criteria

The **Prostate Cancer Clinical Trials Working Group 3 (PCWG3)** clinical states model classifies men by prior therapy, from rising PSA through non-metastatic (nmCRPC) to metastatic CRPC (mCRPC).

- PCWG3 PSA progression is a rise of ≥25% and ≥2 ng/mL above nadir, confirmed ≥3 weeks later.

Modality	Pitfall	PCWG3 rule
PSA	Early rise with chemotherapy; flare with radium-223 and ^{177}Lu-PSMA	Confirm any rise
Bone scan	Early new lesions may be **flare** of healing disease	Rescan at **12–16 weeks** as the new baseline; progression needs **≥2 new lesions** confirmed by ≥2 more ("2 + 2")
CT	Soft tissue is little of the burden	RECIST; PFS preferred over response rate
PSMA PET	Apparent progression at 4 weeks resolves by 3 months in ≈13%	Criteria not defined

> **KEY POINT**
> Every marker (PSA, bone scan, PSMA PET) can flare early, so call progression only on confirmed, combined evidence.

- **Continue ADT indefinitely** during every later therapy.

3 · Germline and somatic testing

- **Germline testing** suits all metastatic disease, relevant family history (BRCA, Lynch, HOXB13) and Ashkenazi ancestry.

- **Somatic** sequencing usually uses the archival primary. In liquid biopsy, **clonal haematopoiesis (CHIP)** can give false DNA-repair calls.
- Somatic homologous recombination repair (HRR) mutations occur in **20–25%** of mCRPC. Mismatch repair deficiency (dMMR/MSI-H) occurs in **2–5%** and should prompt Lynch counselling.

4 · Non-metastatic CRPC (nmCRPC)

nmCRPC is a rising PSA on castration with no metastases on **conventional** imaging. PSA kinetics are the strongest predictor. **Add an ARPI (apalutamide, enzalutamide or darolutamide) to ADT when PSADT is ≤10 months.** All three improve metastasis-free survival (MFS) and OS. Abiraterone is active but **not approved**.

5 · mCRPC: androgen-signalling inhibitors

Agent	Mechanism	Dose	Toxicity / notes
Abiraterone	Selective CYP17 inhibitor (17α-hydroxylase and 17,20-lyase)	1000 mg **fasting** with prednisone (10 mg/day in mCRPC)	Mineralocorticoid excess, hepatotoxicity; correct potassium first. Food raises absorption up to 10× (alternative: 250 mg with a low-fat meal)
Enzalutamide	Pure AR antagonist; blocks translocation and DNA binding	160 mg, no steroid	**Seizures (≈1%)**, fatigue, falls; CYP3A4/2C8 interactions
Apalutamide	Does not cross the blood–brain barrier (BBB)	240 mg	**Rash, hypothyroidism**, fractures
Darolutamide	Structurally distinct; does not cross the BBB	600 mg twice daily	Least CNS toxicity

KEY POINT

Prednisone with abiraterone: blocking CYP17 lowers cortisol, so ACTH rises and causes mineralocorticoid excess; prednisone suppresses ACTH. Enzalutamide needs no steroid. All ARPIs cause falls and fractures, so take care over 75.

6 · Cytotoxic chemotherapy

Cabazitaxel has poor P-glycoprotein affinity, so it works after docetaxel.

Agent	Dose / line	Signature toxicity
Docetaxel	**75 mg/m² 3-weekly** with prednisone 5 mg twice daily; first line	Neutropenia, **neuropathy**, oedema, hyperlacrimation, nail dystrophy
Cabazitaxel	**20 mg/m²** (25 if low neutropenic risk) 3-weekly with prednisone and G-CSF; after docetaxel	**Neutropenia** (grade 3–4 82%), **diarrhoea**
Mitoxantrone	12 mg/m² with prednisone; palliation	No survival benefit
Platinum (with etoposide or taxane)	Neuroendocrine variant; DNA-repair-deficient	—

7 · Radiopharmaceuticals

Feature	Radium-223	^{177}Lu-PSMA-617
Emitter	**Alpha** (path **<100 µm**, so marrow is spared); calcium mimetic	Beta; needs PSMA PET-positive disease
Eligible	**Symptomatic bone metastases, no visceral or bulky nodal disease**	PSMA-avid, any site, after ARPI and taxane
Schedule	50 kBq/kg IV 4-weekly × 6	6-weekly, up to 6 cycles
Toxicity	Diarrhoea, anaemia	Fatigue, **dry mouth**, anaemia
Caution	**Never with abiraterone/prednisone**; bone agent with any ARPI	—

Strontium-89 and samarium-153 only palliate pain.

8 · PARP inhibitors

P4 PARP inhibitors in Prostate cancer — synthetic lethality, PARP trapping, AR–PARP synergy, BRCA2 vs ATM, approvals, toxicity, key trials

Not in the paper answer:

- PARPIs are ineffective in unselected men; haemoglobin can fall suddenly, so watch blood counts.

9 · Immunotherapy

- **Sipuleucel-T**, an autologous dendritic-cell vaccine, was the first approved therapeutic cancer vaccine.
- It suits **asymptomatic/minimally symptomatic mCRPC with no visceral metastases or opioid pain**. It prolongs OS **without** PSA or PFS change.
- **Pembrolizumab** 200 mg IV 3-weekly is approved for **MSI-H/dMMR** or tumour mutational burden **≥10 mutations/megabase** after ≥1 line.

10 · Neuroendocrine/aggressive-variant CRPC

Feature	Usual mCRPC	Neuroendocrine variant
AR / PSA	Retained; PSA tracks disease	Lost; **PSA low despite progression**
Pattern	Osteoblastic bone ± nodes; visceral <20%	Bulky masses, liver, **osteolytic** lesions with hypercalcaemia, brain
Molecular	AR alterations	≥2 of PTEN loss, TP53 mutation, RB1 loss; low PSMA
Treatment	ARPI, taxane, radioligand, PARPI	**Platinum + etoposide** (or taxane + carboplatin) ± radiotherapy (RT)
Survival	Years	**<12 months**

Biopsy a metastasis if progression is rapid with a low PSA.

11 · Sequencing and cross-resistance

A second ARPI helps only **15–30%** because of cross-resistance, so **switch mechanism** (CARD). See the P2-140 sequencing table; add pembrolizumab for MSI-H/dMMR and a PARPI if HRR-mutated.

- Enzalutamide induces CYP3A4/2C9/2C19, so check anticoagulants and antiepileptics.
- AR-V7 in circulating tumour cells predicts resistance to abiraterone and enzalutamide, not to taxanes.

12 · Complications and palliative care

P2 Metastatic prostate cancer with bone metastases — cord compression pathway, Patchell, pain RT, obstruction, zoledronic acid vs denosumab, ONJ prevention

Not in the paper answer:

- Obstructive uropathy must be **relieved before chemotherapy**, but not every unilateral case needs drainage.

Every focal back pain in a man with bone metastases is cord compression until proven otherwise.

> **ALGORITHM**
>
> 1. Back pain with or without deficit or sphincter change: suspect compression.
> 2. Give **high-dose IV dexamethasone** (10 mg IV, then 4–10 mg 6–12-hourly; 16–100 mg/day).
> 3. Arrange **urgent whole-spine MRI**.
> 4. Give RT with or without decompressive surgery.
> 5. Taper steroids over 2–3 weeks.

- Operate for progression during RT, unstable fracture, or recurrence after RT.
- Treat neuropathic pain with tricyclics or gabapentinoids, and start hospice care in time. Erythropoiesis-stimulating agents are not recommended for anaemia.

13 · Bone-targeted therapy

Two uses: ADT osteoporosis, and **prevention of skeletal-related events (SREs)** in CRPC bone metastases. There is **no survival effect**.

Feature	Zoledronic acid	Denosumab
Dose	**4 mg IV 4-weekly**, renal adjustment	**120 mg subcutaneous (SC) 4-weekly**; no renal adjustment
SRE	Better than placebo	**Better than zoledronic acid** (20.7 vs 17.1 months)
Toxicity	Flu-like, creatinine rise, hypocalcaemia, osteonecrosis of the jaw (ONJ)	Hypocalcaemia (grade ≥3 5%), ONJ **2–4%**

> **KEY POINT**
>
> **ONJ** follows dental extraction or poor teeth, so check the teeth first. Give calcium 1000 mg and vitamin D 800 IU daily with both agents. Bone agents are strongly recommended with radium-223.

14 · Landmark trials

Trial	Design	Result	Take-home
SPARTAN	nmCRPC, PSADT ≤10 mo; apalutamide	MFS 40.5 vs 16.2 mo (HR 0.28); OS HR 0.78	First nmCRPC drug

Trial	Design	Result	Take-home
PROSPER	nmCRPC; enzalutamide	MFS improved; death reduced 27%	Enzalutamide for nmCRPC
ARAMIS	nmCRPC; darolutamide	MFS 40.4 vs 18.4 mo; OS HR 0.69	Least toxic
TAX 327	Docetaxel 3-weekly vs weekly vs mitoxantrone	OS 18.9 vs 17.3 vs 16.4 mo (HR 0.76)	First chemo to prolong survival
TROPIC	Post-docetaxel; cabazitaxel vs mitoxantrone	OS 15.1 vs 12.7 mo (HR 0.70)	Second-line chemo
PROSELICA / FIRSTANA	Cabazitaxel 20 vs 25 / vs docetaxel first line	20 non-inferior / not superior	20 mg/m²; docetaxel first
CARD	After docetaxel and 1 ARPI: cabazitaxel vs 2nd ARPI	OS 13.6 vs 11.0 mo (HR 0.64)	Switch mechanism
COU-AA-301	Post-docetaxel; abiraterone	OS 14.8 vs 10.9 mo (HR 0.65)	Post-chemo
COU-AA-302	Chemo-naive, asymptomatic	rPFS 16.5 vs 8.3 mo	Pre-chemo
AFFIRM	Post-docetaxel; enzalutamide	OS 18.4 vs 13.6 mo (HR 0.63)	Post-chemo
PREVAIL	Chemo-naive; enzalutamide	Death reduced 29%; radiographic progression reduced 81%	Pre-chemo
ALSYMPCA	Symptomatic bone-only; radium-223	OS 14.9 vs 11.3 mo (HR 0.70); SRE delayed	Alpha emitter prolongs life
VISION	Post-ARPI and taxane, PSMA-positive; ^{177}Lu-PSMA	OS 15.3 vs 11.3 mo (HR 0.62); rPFS HR 0.40	Approved 2022
TheraP	Phase II; ^{177}Lu-PSMA vs cabazitaxel	PSA response 65 vs 37%; fewer grade 3–4; OS equal	Lutetium ≥ cabazitaxel
IMPACT	Sipuleucel-T vs placebo	OS 25.8 vs 21.7 mo (HR 0.78); no PSA/PFS change	Vaccine approved 2010
PROfound	Post-ARPI; olaparib vs 2nd ARPI	BRCA/ATM rPFS 7.4 vs 3.6 mo (HR 0.34)	Olaparib, HRR panel
TRITON2/3	Rucaparib after ARPI	BRCA rPFS HR 0.50; no ATM benefit	BRCA only
PROpel / MAGNITUDE / TALAPRO-2	First-line PARPI with abiraterone / with enzalutamide	rPFS gains, largest in BRCA; PROpel OS 42.1 vs 34.7 mo (P = 0.054)	HRR/BRCA
Denosumab vs zoledronic acid	Bone-metastatic CRPC	First SRE 20.7 vs 17.1 mo; OS equal	Denosumab for SRE

★ Putting it together

A · PSA to diagnosis

ALGORITHM

1. **Raised prostate-specific antigen (PSA) or abnormal DRE**: repeat it; exclude infection and instrumentation; double the PSA on 5α-reductase inhibitors (→ Ch 154).
2. **Multiparametric MRI (mpMRI) before biopsy**: PI-RADS (→ Ch 155).
3. **Biopsy** PI-RADS 3–5 targets plus systematic cores, preferably transperineal.
4. Assign **ISUP Grade Group (GG)** (→ Ch 156) and risk group (→ Ch 157). If unfavourable intermediate or worse, stage with PSMA PET (or CT and bone scan).

B · Localised disease: matching risk to treatment

NCCN favourable groups need ALL criteria, adverse groups ANY one.

Risk group	Defined by	Management
Very low	T1c, GG1, PSA <10, density <0.15, <3 positive cores (≤50% each)	Active surveillance (AS)
Low	T1–T2a, GG1, PSA <10	AS preferred
Favourable intermediate	One factor (T2b–c, GG2, PSA 10–20); <50% cores	Radical prostatectomy (RP) or radiotherapy (RT); AS if low-volume GG2
Unfavourable intermediate	2–3 factors, or GG3, or ≥50% cores	RP, or RT with 4–6 months ADT
High / very high	T3a, GG4–5 or PSA >20 / T3b–T4, primary pattern 5, ≥2 high-risk features	RP with extended node dissection, or RT with ADT 18–36 months; add abiraterone 2 years if N1 or ≥2 of Gleason ≥8, cT3–4, PSA ≥40 (STAMPEDE)

KEY POINT

SPCG-4, PIVOT and ProtecT show that radical treatment helps only younger men with ≥intermediate-risk disease. ProtecT: RP = RT for 15-year cancer survival.

C · Surgery vs radiotherapy vs focal therapy

	RP (Ch 160–161)	RT (Ch 162)	Focal (Ch 163)
Main toxicity	Incontinence, ED	Rectal/urinary irritation	Least; retreatment common
Advantages	Pathological staging	Suits comorbid men	Preserves function
Failure	PSA ≥0.2 ng/mL (confirmed)	Phoenix: nadir + 2 ng/mL	Undefined; biopsy-based
Salvage	Bed RT early at PSA 0.1–0.2 ± ADT	RP, cryotherapy, HIFU, brachytherapy	Repeat focal, RP, RT

D · Advanced disease

ALGORITHM

1. **Biochemical recurrence** (Ch 165): PSMA PET. After RP, salvage RT; after RT, local salvage if biopsy-proven. If high risk (PSADT ≤9 months), give ADT and enzalutamide (EMBARK).
2. **mHSPC**: never ADT alone; add an ARPI, or a docetaxel triplet if fit and high volume. If low volume, add prostate RT (STAMPEDE arm H). Test genes in all.
3. **nmCRPC** with PSADT ≤10 months: ARPI (SPARTAN, PROSPER, ARAMIS).
4. **mCRPC** (Ch 166): switch mechanism. Docetaxel → cabazitaxel (CARD); ^{177}Lu-PSMA (VISION); radium-223 for symptomatic bone-only disease (ALSYMPCA); PARP inhibitor if BRCA/HRR mutation (PROfound); pembrolizumab if MSI-H; protect bone.